

Neurology

Mina Atia MD FRCPC

Saturday, Nov 29th 2025



"Our brains are similar to computers. Your husband's brain is like an old computer. It locks up, lacks memory and is filled with useless stuff."

Outline

1. Stroke

2. MS

3. Movement Disorders

- Tremor
- Idiopathic Parkinson disease
- Parkinson Plus syndromes

4. Peripheral Neuropathy

5. ALS (if time)

Stay tuned for your **Neuro Emergencies Lecture:**
(Sunday Nov 30th)

- seizures, headaches, NMJ disorders like
Myasthenia Gravis, GBS, Cord syndromes

Bonus Videos on internalmedicineinreview.ca

- Neuroanatomy 101
- Neuro-ophthalmology
- Examine them for... Root vs Nerve
- Cranial nerve exam
- Stroke syndrome exams

Bonus Slides:

Neuro-ophthalmology, Neuromuscular (Root vs Nerve), LP, Approach to CT/MRI Imaging, ALS, COVID

Stroke

1. Hyperacute management of ISCHEMIC stroke (in the ER or ICU)

- Ischemic Stroke: Thrombolysis, Antiplatelets, BP Targets, Seizure management etc.
- Hemorrhagic Stroke: Hyperacute management of ICH

2. Secondary Workup of Stroke or TIA

3. Management by Mechanism and Secondary Stroke Prevention

- Cardioembolic: AFIB, PFO, ESUS
- Atheroembolic: Carotid stenosis, Dissection
- Lacunar and other “Garden variety” stroke secondary prevention

4. Stroke in Special Populations: Young or Pregnant, Cerebral Venous Thrombosis (CVT = **NEW in 2024!**), Cancer associated Stroke

5. Stroke Syndromes to Recognize for your exam



MCQ #1 2026

A 73F was well 3 hours ago but then developed profound right sided weakness, left gaze deviation and dysarthria. She has a right homonymous upper quadrantonopia on exam. CT head shows extensive hypodensity in the left parietal, temporal, and frontal lobes (MCA and ACA). CTA shows occlusion of the left distal carotid, proximal MCA. What is the best next step in management?

- a) ASA 162mg
- b) ASA 162mg and Clopidogrel 300mg
- c) IV TNK
- d) IV alteplase
- e) Endovascular Thrombectomy

Hyperacute Management of Ischemic Stroke

Investigations

Stroke Symptoms <24h = “CODE STROKE”

- Immediate evaluation for IV thrombolysis (tPA or tNK) or endovascular therapy (EVT)
- ABCs + NIHSS + evaluate/treat seizures if present
Note: Seizure in the presence of suspected acute stroke is not a contraindication for revascularization and could be treated using appropriate short-acting medications (e.g., lorazepam IV) if the seizures are not self-limited
- STAT Bloodwork: CBC, lytes, glucose, INR, PTT, Cr + ECG

STAT Neuroimaging

- Onset <4.5h = CT Head to determine if tPA/tNK candidate
- Onset <6h = CT, CTA arch to vertex to determine if EVT candidate
- Onset >6 but <24h and eligible for *late window EVT* = CT, CTA, CTP*
*CTP = CT Perfusion - determines if late EVT is an option

Hyperacute Management of Ischemic Stroke

Thrombolysis

Thrombolysis Inclusion Criteria

- Ischemic stroke causing **disabling** neurologic deficit in patient >18yo
- Time from last known well (onset of stroke symptoms) **<4.5h**

Absolute Exclusion Criteria

- Any source of active hemorrhage or any condition that could increase the risk of major hemorrhage with thrombolysis*

***if on DOAC, cannot give tPA, but can consider EVT**

- Any hemorrhage on brain imaging

Relative Exclusion Criteria

- See strokebestpractices.ca, + next slide

BONUS
Read on own

BONUS: READ ON YOUR OWN

Thrombolysis (tPA or TNK) exclusions

Note in training most of you have worked at a centre with access to a Stroke Neurologist for decisions of thrombolysis... on exam (and in life) YOU may be the expert so need to know these!

Hint for
MCQ1

Absolute Exclusion Criteria

- Any source of active hemorrhage or any condition that could increase the risk of major hemorrhage after intravenous thrombolysis administration.
- Any hemorrhage on brain imaging.

Relative Exclusion Criteria (requiring clinical judgement based upon the specific situation. Consult Stroke Specialist at Comprehensive Stroke Centre if there are any questions or concerns about these criteria).

Historical

- History of intracranial hemorrhage.
- Stroke or serious head or spinal trauma in the preceding 3 months.
- Major surgery (e.g., cardiac, thoracic, abdominal, or orthopedic) in the preceding 14 days. Risk varies according to the procedure.
- Arterial puncture at a non-compressible site in the previous 7 days.

Clinical

- Stroke symptoms due to another non-ischemic acute neurological condition such as seizure with post-ictal Todd's paralysis or focal neurological signs due to severe hypo- or hyperglycemia.
- Hypertension refractory to aggressive hyperacute antihypertensive treatment such that target blood pressure <180/105 cannot be both achieved and maintained.
- Currently prescribed and taking a direct non-vitamin K oral anticoagulant. [Refer to Section 5.2](#)

CT or MRI Findings

- CT showing early signs of extensive infarction (e.g., >1/3 of middle cerebral artery [MCA] territory, or ASPECTS score <6).

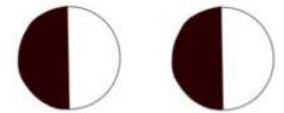
Laboratory

- Blood glucose concentration <2.7 mmol/L or >22.2 mmol/L.
- Elevated activated partial-thromboplastin time.
- International Normalized Ratio >1.7.
- Platelet count <100,000 per cubic millimetre.

Clarifying “disabling” stroke & 4.5h

Thrombolysis Inclusion Criteria

- Ischemic stroke causing **disabling** neurologic deficit in patient >18yo
- Disabling is a judgment call but in general:
 - disabling = NIHSS ≥ 6
- OR
- disabling = aphasia, hemianopia, weakness limiting sustained effort against gravity, visual/sensory extinction
- Time from last known well (onset of stroke symptoms) <4.5h



NIHSS = only 2
...But very disabling

Note: ARAMIS trial JAMA 2023 showed dual antiplatelet is non-inferior to tPA in *non-disabling/minor strokes*

Thrombolysis Agent (tPA vs tNK)

Tenecteplase (tNK) may be considered as an alternative to alteplase (tPA) within 4.5 hours of acute stroke symptom onset

Tenecteplase (tNK) dose: 0.25 mg/kg up to a **maximum of 25mg**** **given as a single bolus over 5 seconds**

****vs STEMI or acute massive PE - 0.5mg/kg up to 50mg****

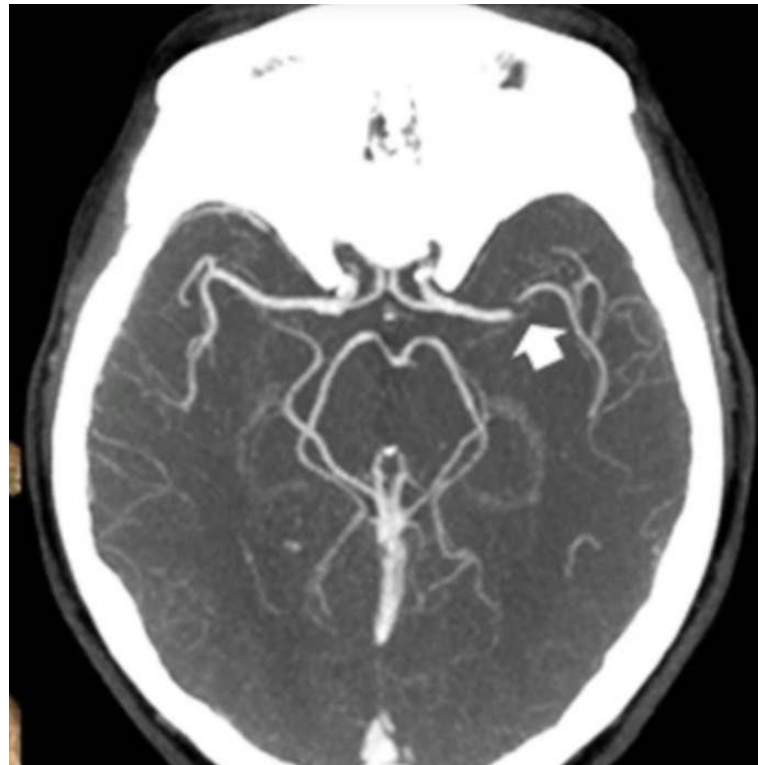
Note: Alteplase (tPA) dose is 0.9mg/kg, 10% (0.09mg/kg) given as IV bolus over 1 min and 90% (0.81mg/kg) **given as IV infusion over 60min**



(ER Nurses strongly prefer TNK for ease of administration - does not require programming rapid infusing pump!)



70yF R-handed with acute onset right sided weakness
and aphasia 2 hrs ago...



Hyperacute Management of Ischemic Stroke

Endovascular Thrombectomy (EVT)

EVT Inclusion Criteria

Consider EVT (in addition to thrombolysis or in pts not eligible for thrombolysis) if

- ✓ >18yo w/ disabling stroke AND functionally independent* and life expectancy >3m
- ✓ <6h** from onset of stroke or last known well
- ✓ CT Head shows small-to-moderate ischemic core (ASPECTS score of 6 or higher)
- ✓ CTA shows occlusion in the anterior circulation of proximal large vessel (distal ICA or MCA)
- ❖ no evidence for EVT in proximal posterior circulation stroke, but can consider it for basilar thrombus because of high morbidity/mortality

*the use of intravenous thrombolysis and/or EVT in patients who are *not functionally independent* may be considered, based on careful review of risks and benefits for the patient

**for patients presenting between 6 and 24h from last seen well, highly selected patients may receive EVT if they meet clinical and imaging criteria (beyond scope of GIM exam to know criteria, but DO “consult stroke specialist / EVT centre” in oral scenario)

Hyperacute Management of Ischemic Stroke

Blood Pressure Targets first 24h

Scenario	BP Target	COMMENTS
Thrombolysis being given	<180/105 x 24 h	185/110 vs 180/105?
Thrombolysis + EVT	<180/105 x 24 h	Guideline states for 24h post tPA target BP <180/105 so in order to give thrombolysis In ED target <180/105
EVT only	? (<220/120?)	No target in stroke guidelines ESCAPE Trial appendix <220/120 “whatever the interventionalist wants”
No thrombolysis or EVT	<220/120	Permissive hypertension first 24h
Post Thrombolysis Hemorrhage	<185/110	Specific target and duration of therapy unknown

Hyperacute Management of Ischemic Stroke

Secondary ICH management

Post tPA/tNK: Consider Intracerebral Hemorrhage (ICH) if change in neuro status e.g. decreased LOC, spike in BP, new/worse headache → Stat CT head

- if no ICH, consider CTA to look for vessel occlusion —> consider EVT
 - if ICH:
 - ✓ stop IV thrombolysis if infusing
 - ✓ stat CBC, INR, type and cross
 - ✓ If patient hypertensive >185/110: TREAT, however specific target and duration of therapy are unknown
 - ✓ consider cryoprecipitate, human fibrinogen concentrate, fresh frozen plasma, tranexamic acid (potential benefit, limited harm, case by case basis)
- Do not use prothrombin complex concentrates, platelet transfusions, factor VIIa (no benefit, potential harm)

BONUS
Read on own

Stroke in the ER - Acute Stroke – post thrombolysis bleeding

Post-tPA/tNK systemic bleeding

- Consider systemic bleeding if visible bleeding, reduced BP, localized pain, diaphoresis
- if systemic (extracranial) bleeding
 - stop IV thrombolysis if infusing
 - compress and use ice if possible if puncture site hematoma
 - stat CBC, INR, fibrinogen
 - transfuse as per usual protocols
 - CT triphasic abdo if back pain /concern for retroperitoneal bleed

Hyperacute Management of Ischemic Stroke

Antiplatelets



- All acute ischemic stroke + not on antiplatelet already + no thrombolysis = **Give ASA 160mg** (or sometimes Dual Antiplatelet)

Note: antiplatelet or dual does not preclude EVT

- if given thrombolysis = **wait 24h before starting ASA** (no dual antiplatelet)
- if already on an antiplatelet: individualize decision making
“Continuing the current antiplatelet agent or switching to a different agent are reasonable options. At the time of writing, evidence is lacking to make more specific recommendations”

Acute Management of Ischemic Stroke

Antiplatelets

For high risk TIA (ABCD² ≥4) or minor stroke (NIHSS ≤3) of non-cardioembolic origin, not at high bleeding risk:

- **Short-term dual antiplatelets** are recommended
 - ASA 162mg + clopidogrel 300-600mg loading dose followed by ASA 81mg + clopidogrel 75 mg daily, **for 21 days** then antiplatelet monotherapy
- OR**
- ASA 162 + ticagrelor 180 mg loading dose followed by ASA 81mg OD + ticagrelor 90 mg BID **for 30 days then ASA monotherapy**
 - start ASAP after brain imaging, within 24 hours of symptom onset, ideally within 12 hours.

Which anti platelet regimen to pick?

No RCT / head to head trials of clopidogrel vs ticagrelor for this indication. Meta-analysis suggests no major differences in outcomes. Guidelines state to individualize based upon patient clinical factors.

Ticagrelor is not currently covered in Ontario for this indication so we don't use it a lot in stroke world (\$)

Acute Management of Ischemic Stroke

Antiplatelet Nuances

Scenario: Pt presenting with TIA/stroke, *already taking* ASA 81mg OD or clopidogrel 75mg OD

- address stroke etiology, manage vascular risk factors
- insufficient evidence to guide management if a stroke occurs while pt on an antiplatelet
 - Either continuing current agent or switching to another antiplatelet are reasonable options
- Platelet function assays or pharmacogenomic testing for clopidogrel resistance exist, but clinical implications for stroke prevention treatment unclear.

Acute Management of Ischemic Stroke

Antiplatelets

Severe **Intracranial** Atherosclerosis

- patients with TIA/stroke in last 30 days and 70-99% stenosis in major intracranial vessel and low bleeding risk
 - = consider **dual antiplatelet (ASA & Clopidogrel) x 3 months**
 - followed by single antiplatelet after 3 mos
 - Plus LDL lowering, BP treatment, structures lifestyle medication (smoking cessation, exercise, diet)

**** This is why you want CTA, not Carotid Doppler Ultrasound (CDUS) in your diagnostic workup!**



BONUS
Read on own

More on Switching Antiplatelets

Should you switch to clopidogrel if stroke occurred on ASA?

There is no RCT to answer this specific question, but in CAPRIE (1996) – randomized patients with acute stroke to either ASA vs clopidogrel

- Clopidogrel > ASA in secondary prevention of composite outcome of stroke/MI/vasc death in pts w/ prior MI/stroke/PAD

RRR ~9%

ARR ~0.5%

NNT 200

= Thus, very little evidence, very little benefit, though it is common practice to switch

Note – very old study guides may reference Aggrenox (ASA+dipyridamole), this is no longer in stroke best practice guidelines for use.

! Acute Management of Ischemic Stroke: Antiplatelet Summary Slide

Scenario – ISCHEMIC	Regimen	NOTES
Thrombolysis given	Delay Antiplatelet 24h	ASA 81mg after 24h
No Thrombolysis, but EVT	ASA 160mg loading immediately	Screen for dysphagia, if present either ASA 81mg per NG or 325mg rectally
No Thrombolysis Major stroke (NIHSS ≥ 4)	ASA 160mg loading immediately	
No Thrombolysis High risk TIA $ABCD^2 \geq 4$ or Minor stroke (NIHSS ≤ 3)	ASA 162mg + clopidogrel 300-600mg loading $\rightarrow$ ASA 81 + clopidogrel 75 x 21d $\rightarrow$ ASA or clopidogrel indefinitely OR ASA 162mg + ticagrelor 180mg loading $\rightarrow$ ASA 81+ticagrelor 90 bid x 30d $\rightarrow$ ASA indefinitely	Exclusion for 2 nd antiplatelet: Hx ICH in past -bleeding diathesis, plt <100 -surgery within 30d -GI bleed within 6 mos -hx Afib or suspicion of cardioembolic cause -already on DOAC/VKA -ESRD or ESLD
Patient already on an antiplatelet at time of stroke	? Choose your own strategy “Continuing the current antiplatelet agent or switching to a different agent are reasonable options. At the time of writing, evidence is lacking to make more specific recommendations.”	



Patient in AFIB or carotid/intracranial stenosis = Discussed separately

MCQ #1 2026

A 73F was well 3 hours ago but then developed profound right sided weakness, left gaze deviation and dysarthria. She has a right homonymous upper quadrantonopia on exam. CT head shows extensive hypodensity in the left parietal, temporal, and frontal lobes (MCA and ACA). CTA shows occlusion of the left distal carotid, proximal MCA. What is the best next step in management?

- a) **ASA 162mg**
- b) ASA 162mg and Clopidogrel 300mg
- c) IV TNK
- d) IV alteplase
- e) Endovascular Thrombectomy

Answer: A

Stroke is already very large and established, thus thrombolysis with either TNK or alteplase and Thrombectomy are not options despite being within the time window. No dual antiplatelet as large stroke.



Post-Stroke Seizure Management

- Treat new-onset seizure acutely with **short acting benzodiazepine** if seizure is not self limited
- “Immediate post-stroke seizure” does not require long-term anticonvulsant medications [**Conditional recommendation**; Low quality of evidence]
 - Immediate post-stroke seizure = single sz, self-limiting, within 24h of stroke onset
- No role for prophylactic anticonvulsants in acute stroke



Post-Stroke Seizure Management

What we stroke-ologists do practically...

- Post stroke seizure lasting >2min [“not self-limited”]
 - treat as status epilepticus as you normally would, IV lorazepam etc [see Neuro emergencies lecture]
- Seizure, not self-limited occurring **<2wks** from time of stroke
 - use AED for ~3 months then reassess and consider stopping
- Seizure occurring **>2wks** from time of stroke = late post stroke → use AED long term

Intracranial Hemorrhage (ICH)

2020 Update on ICH Management

Identify cause

Treat coagulopathy

Treat Blood pressure

Prevent complications

- CTA indicated for most to rule out underlying lesion (aneurysm/avm/venous thrombosis causing ICH)



CTA is better than MRA to detect vascular lesions



Intracranial Hemorrhage (ICH)

2020 Update on ICH Management

ICH on Anticoagulant

- Reverse Anti-Xa (apixa/edoxa/riva) with PCC (50u/kg up to 3000u)
- Reverse dabigatran with idarucizumab 5g IV; if unavailable FEIBA (activated PCC) 50u/kg up to 2000u
- LMWH therapeutic within 12h - consider protamine
- IV uFH - protamine
- Antiplatelet - n/a (do not give platelets)
- Tranexamic acid - may ↓ volume of ICH but doesn't improve 90d outcomes (TICH trial)
- When to restart anticoagulants: individualize decision per patient

Intracranial Hemorrhage (ICH)

2020 Update on ICH Management

Acute BP Target

- Assess BP q15min until target achieved for first 24h
- sBP <140 does not worsen outcomes compared to <180 but clinical benefit not established
- **sBP <140-160mmHg for first 24-48h may be reasonable**
 - <140 target favoured if onset <6h ago, anticoag therapy, presence of markers of ICH expansion, presenting SBP <220mmHg

Long Term BP Target

- **<130/80** - use home monitors to maintain this where possible

Intracranial Hemorrhage (ICH) on anticoagulant

AHA/ASA 2022 Guideline on Management of Spontaneous ICH

- In a pt with afib and ICH, when to restart anticoagulation?
 - ~7-8 weeks after ICH
- In a pt with afib in whom you deem it is not safe to ever restart anticoag (e.g. evidence of cerebral amyloid angiopathy)
 - can consider atrial appendage closure

Intracranial Hemorrhage (ICH)

2020 Update on ICH Management

Role of neurosurgeon

- **on your exam: ALWAYS call neurosurg for ICH (and generally in real life too)**
- external ventricular drainage should be considered if ↓ LOC and hydrocephalus are present
- surgery is NOT recommended if patient stable and no signs of herniation.

Workup for cause of ICH

- Consider MRI to look for cerebral amyloid angiopathy/mass/AVM/AVF
 - Timing of MRI unclear – acute blood obscures underlying pathology

Other housekeeping:

- DVT prophylaxis: Intermittent pneumatic compression devices initially, then LMWH at 48h after documentation of stabilization of hematoma (eg repeat CT head sometime after 24h of admission)
- Acute seizure in first 24h - do not need longterm AED (short term AED is an option).
 - No role for prophylactic AED with de novo ICH.
- Use of steroids to treat ICP with ICH may cause harm

Stroke

1. Hyperacute management of ISCHEMIC stroke (in the ER or ICU)
 - Thrombolysis, Antiplatelets, BP Targets, Seizure management and more...
 - Hyperacute management of HEMORRHAGIC stroke
2. **Secondary Workup of stroke or TIA**
3. Management by Mechanism and Secondary Stroke Prevention
 - Cardio-Embolic: AFIB, PFO, ESUS
 - Athero-Embolic: Carotid stenosis, Dissection
 - Lacunar and other “Garden variety” stroke secondary prevention
4. Stroke in Special Populations: Young or Pregnant, Cerebral Venous Thrombosis (CVT = **NEW in 2024!**), Cancer associated Stroke
5. Stroke Syndromes to Recognize for your exam



Outpt Workup of TIA/non-disabling stroke

Pt with TIA/stroke presenting after 24h to outpatient setting

- CT Head or MRI Head
- ideally CTA (or MRA) aortic arch to vertex
 - carotid dopplers (CDUS) are also listed as option in guideline, but do not tell you about intracranial disease and not useful in posterior circulation stroke
- Bloodwork: CBC, lytes, PTT, INR, Cr, ALT, Glucose, Lipid Profile, HbA1c
 - Consider BHCG in females less than 50yo
 - If giant cell arteritis suspected (e.g., age >50, ischemic optic neuritis / retinal ischemia or headache), check ESR and CRP
 - Extensive thrombophilia testing for hereditary hypercoagulable disorders is **not** recommended for routine investigation of a patient with ischemic stroke and should be limited to selected situations (e.g. unexplained cerebral venous thrombosis; PFO-related paradoxical embolism).
 - *If thrombophilia is suspected, recommend consult with stroke or thrombosis expert ... **Do not do APLAS first line, rule out AFIB etc first***
- ECG+ Holter ≥24h
 - Note: if suspect cardioembolic stroke = 2 week holter
- ECHO if suspect stroke is of embolic etiology or TIA of uncertain cause

ECHO and Stroke

Pt with TIA/stroke presenting after 24h to outpatient setting

Routine ECHO is NOT required for all stroke patients

- ✓ ECHO should be done if stroke is thought to be embolic or TIA is of unclear cause
- ✓ For pt <60yo who have ?embolic stroke or TIA of unclear cause, ECHO with saline bubbles is recommended to look for PFO
 - ✓ Contrast-enhanced (agitated saline) transesophageal echocardiography (TEE) or transcranial Doppler has greater sensitivity than transthoracic echocardiography for detection of right-to-left cardiac and extra-cardiac shunts

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After the Acute Stroke – Atrial Fibrillation

Management Options

- DOACs are favoured over warfarin for majority (Apixaban, Rivaroxaban, Dabigatran, Edoxaban)
 - DOACs, monitor renal function q12m (CCS 2020 says q6m)
 - Warfarin, monitor INR, keep b/w 2-3
- Bridge with antiplatelet NOT with heparin
- Mechanical Valve or mod/severe Mitral stenosis (valvular afib): warfarin is the only option (again, bridge w antiplatelet, not heparin in almost all cases)

After the Acute Stroke – Atrial Fibrillation

Timing of anticoagulation initiation post stroke

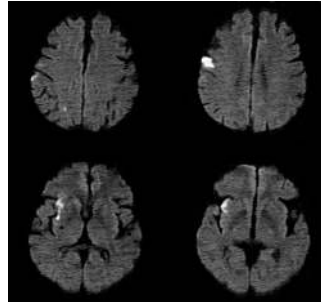
- Expert consensus in guidelines

- TIA = <1 day

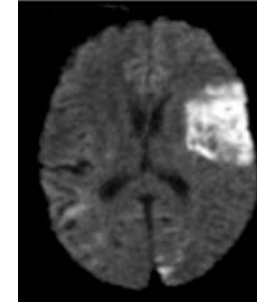
Stroke:

- mild/small = 3 days
- moderate = 6 days
- severe/large = 12 days

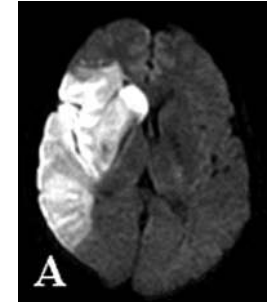
small



moderate



large

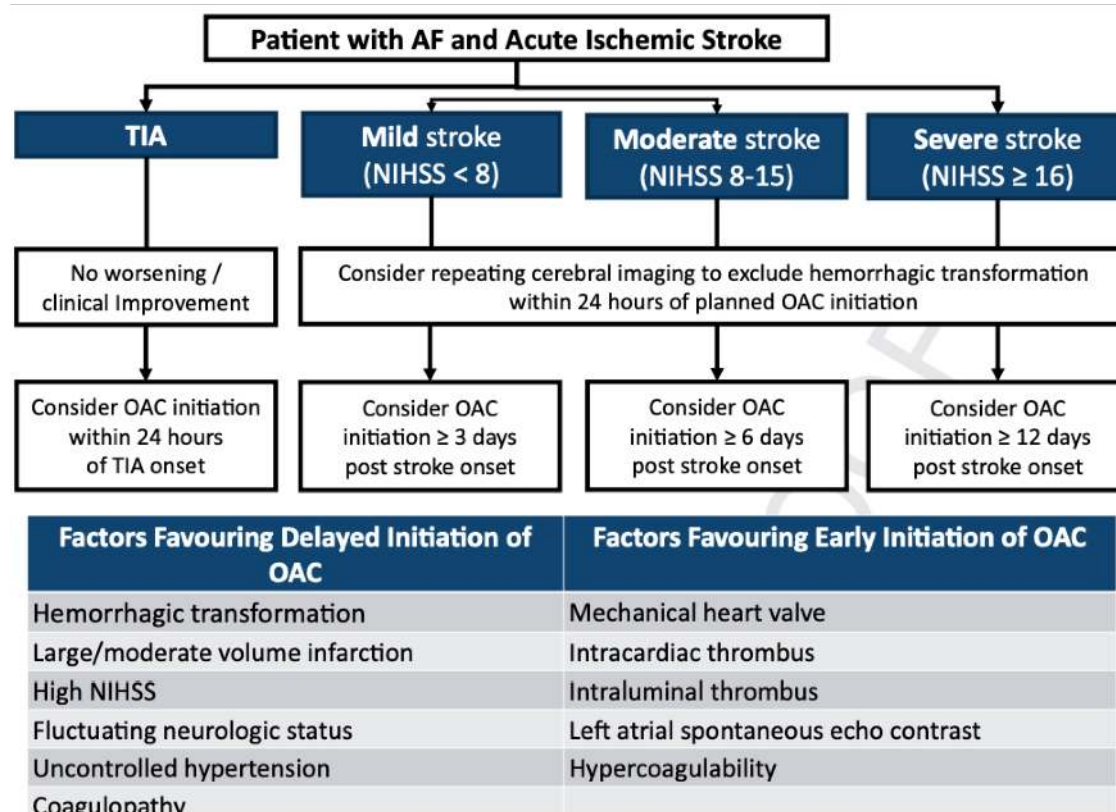


- rpt CT head if Stroke to ensure no bleed prior to initiating anticoagulation

Timing of OAC initiation post Stroke

Timing of anticoagulation after stroke

CCS 2020 Afib guidelines give some guidance for first time



After the Acute Stroke – Atrial Fibrillation

- **Timing of anticoagulation initiation post stroke (NEJM 2023)**

Early versus Later Anticoagulation for Stroke with Atrial Fibrillation

Urs Fischer, M.D., Masatoshi Koga, M.D., Ph.D., Daniel Strbian, Ph.D., Mattia Branca, Ph.D., Stefanie Abend, B.Sc., Sven Trelle, M.D., Maurizio Paciaroni, M.D., Götz Thomalla, M.D., Patrik Michel, M.D., Krassen Nedeltchev, M.D., Leo H. Bonati, M.D., George Ntaios, M.D., et al., for the ELAN Investigators*

- Open label (=any DOAC at clinician discretion) study across 15 countries
- N = 2013 pts with ischemic stroke and nonvalvular AFIB, not on prior OAC assigned to
 - Early anticoag= w/in 48h for mild/moderate stroke or day 6/7 for major stroke
 - Late anticoag= day 3/4 for mild, day 6/7 for mod, day 12/13/14 for major stroke
 - Primary composite outcome of recurrent ischemic stroke, systemic embolism, major extracranial bleeding, symptomatic intracranial hemorrhage, or vascular death within 30 days occurred in
 - 2.9% in early anticoag group & 4.1% in late anticoag group
 - CI -2.84 to 0.47

Stroke/TIA with Afib & already on OAC?

“Continuing current OAC or switching anticoagulants are reasonable options.
Evidence is lacking to make specific recommendations.”

routine addition of ASA to chronic anticoagulant therapy is not recommended because of increased bleeding risk without evidence of benefit and potential for harm unless there is a specific medical indication.

Additional recommendations for pts who stroke while on OAC:

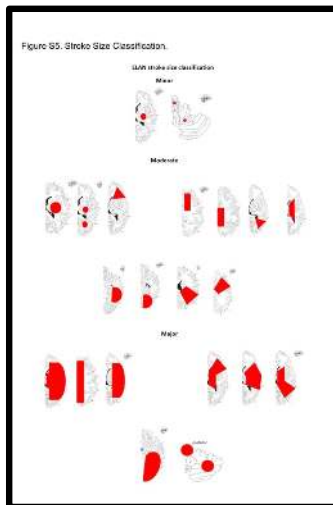
- ✓ address medication nonadherence
- ✓ ensure correct DOAC dosing or warfarin INR control
- ✓ avoid DOAC drug-drug interactions
- ✓ investigate & treat other potential stroke etiology
- ✓ promote general vascular risk factor modification

Stroke/TIA with Afib & already on OAC?

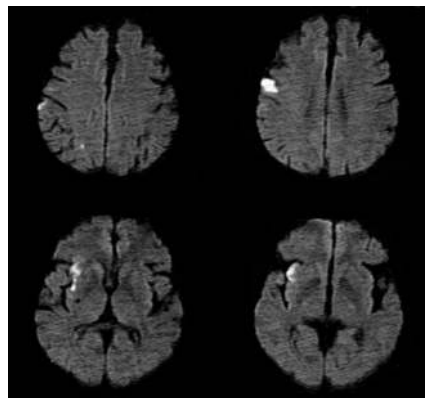
Do you hold the OAC if the acute stroke is moderate or large in size?

This is not in guidelines, but stroke-ologists opinion is...

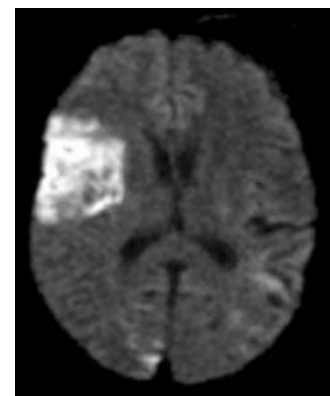
- if moderate/large stroke, reasonable to hold OAC for 6/12 days, use ASA during that time, then rpt CT head and if no bleed, switch ASA to OAC
- if small stroke, reasonable to continue OAC as hemorrhagic risk is lower, but also ok to hold for 3 days, use ASA during that time, then rpt CT head and if no bleed, switch ASA to OAC



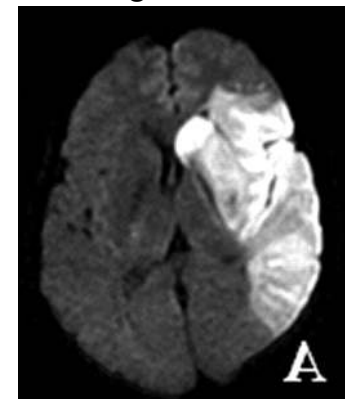
Small



Moderate



Large



After the Acute Stroke - PFO

For carefully-selected pts* with recent stroke attributed to a PFO

PFO closure + anti-platelet > anti-platelet alone if all of the following:

1. **age 18-60***
 2. stroke is non-lacunar by imaging (or pt had cortical symptoms)
 3. stroke specialist thinks PFO is most likely cause after a thorough evaluation to exclude other causes
- “Reasonable” to recommend **against** PFO closure if no high-risk anatomical features
 - High risk features: atrial septal *aneurysm*, large right-to-left shunt (>20 microbubbles), large diameter PFO (≥2mm)

* see slides on stroke in young DDx & work-up

BONUS
Read on own

Self Study – After the Acute Stroke - PFO

FYI only - do not memorize!

Risk of Paradoxical Embolism (ROPE) score for PFO

1. History of HTN - y/n
2. History of DM - y/n
3. History of stroke/TIA - y/n
4. Smoker - y/n
5. Cortical infarct on imaging - y/n
6. Age

FORMULA

Characteristic	Points
No history of hypertension	+1
No history of diabetes	+1
No history of stroke or TIA	+1
Nonsmoker	+1
Cortical infarct on imaging	+1
Age in years	
18-29	+5
30-39	+4
40-49	+3
50-59	+2
60-69	+1
≥70	0

Score interpretation:

RoPE score	PFO-attributable fraction (95% CI)	Estimated stroke/TIA recurrence at 2 years
0-3	0% (0-4)	20% (12-28)
4	38% (25-48)	12% (6-18)
5	34% (21-45)	7% (3-11)
6	62% (54-68)	8% (4-12)
7	72% (66-76)	6% (2-10)
8	84% (79-87)	6% (2-10)
9-10	88% (83-91)	2% (0-4)

Embolic Stroke Undetermined Source

Should patients with ESUS be systemically anticoagulated?

- NAVIGATE (2018)
Rivaroxaban 15mg vs ASA 100mg
No ischemic stroke benefit with Rivaroxaban and increased bleeds
- RE-SPECT ESUS (2019)
Dabigatran 110 or 150mg BID vs ASA 100mg
No difference in ischemic stroke or major bleeds, but more clinically relevant non-major bleeds with Dabigatran
- ATTICUS (**NEJM 2023**)
Apixaban 5mg BID vs ASA 100mg
No difference in ischemic stroke on MRI at 12 months, terminated at interim analysis for futility.

For patients with ESUS, **and no known atrial fibrillation**, anticoagulant therapy is **NOT** currently recommended over ASA 81mg for secondary stroke prevention.

Extracranial Carotid Stenosis (in the neck)

Symptomatic carotid & 50%–99% stenosis should be

- evaluated by stroke expert ASAP, measure stenosis via CTA ideally

In **men with 50%–99% & women with 70%–99%** symptomatic carotid stenosis,

- **Carotid endarterectomy (CEA) is recommended** and should be performed ASAP
- CEA should occur w/in 48h of symptom onset (NNT=3) or within 2wks if patient not stable within first 48h
- CEA is better than Carotid Artery Stenting (CAS) in pts >70yo

In **women with 50 - 69 %** symptomatic carotid stenosis

- CEA may be considered in those at highest risk of stroke recurrence and upon consideration of other patient factors

Note: for pts with mod/severe (disabling) stroke due to symptomatic carotid, benefit of CEA/CAS is uncertain and should be considered on individual basis

Note: Patients with 'tandem lesions' - carotid stenosis + intracranial LVO - many get CAS at time of EVT. For those who did not, consider on case by case basis CEA/CAS in follow up

strokebestpractices.ca

Extracranial Carotid Stenosis (in the neck)

Asymptomatic Carotid:

Asymptomatic or remotely symptomatic (>6m ago) + 60-99% stenosis + life expectancy >5yrs =

- evaluate by stroke expert
- highly selected patients “may” be considered for CEA (or CAS if not CEA candidate)

Extracranial Carotid Stenosis (in the neck)



Summary for your oral exams:

- Symptomatic carotid + 50-99% stenosis = Speak to Vascular Surgeon on call / Stroke Expert about CEA (or CAS if not CEA candidate)
- Symptomatic carotid + <50% stenosis = max medical mgmt
- Asymptomatic carotid stenosis = max medical mgmt

What about posterior circulation stenosis?

- Vertebral artery stenosis: medical management over stenting

Dissection : carotid or vertebral

EXTRAcranial Carotid or Vertebral Dissection

- Give **either anti-platelet or anticoagulant (heparin/warfarin)**
 - Uncertainty about anti-platelet vs anticoagulation (heparin/warfarin) in dissection; either is reasonable per guidelines
 - Lack of evidence regarding duration of treatment
 - Lack of evidence for use of DOACs

INTRAcranial Carotid or Vertebral Dissection

- Lack of evidence for use of anticoagulation in **intra**cranial dissection [give antiplatelet only, bleeding risk is high]

Dissection

CADISS (2015)

- Extracranial carotid or vertebral dissection
- 78% of recruited patients (n=250) presented with TIA/stroke from the dissection (aka symptomatic dissection)
- Randomized to anti-platelet vs anticoagulant (heparin or warfarin)

Results:

- No difference in rate of recurrent stroke
- Rate of recurrent stroke was very low overall

Note: No evidence for DOACs in dissection, thus not used

Note: Never anticoagulate intracranial dissection as risk of bleed is high

Acute Stroke: Dissection

**Summary for your oral exam :
“consult with stroke neurologist”**

+ be safe

(eg) Do not anticoagulate INTRACRANIAL dissection

**(eg) If pt anticoagulated for EXTRACranial dissection w/ acute stroke admit to
monitored bed**

+ think of cause for dissection - Trauma? Aortic dissection?

If atraumatic... Fibromuscular dysplasia if young? Marfan's? Ehler's danlos?

After the “Garden Variety” Stroke

Secondary Prevention



Hypertension/BP Management

- Past stroke/TIA = **BP<140/90***
 - small subcortical stroke = sBP<130 is reasonable
- **Diabetic** patient w/ prior stroke/TIA = BP <130/80

*Remember patients with stroke excluded from SPRINT trial. Stick to stroke guidelines.

Lipid Management

- Must check: LDL, HDL, Total Chol, TG levels
- Start statin : target LDL<1.8
 - if LDL still >1.8 with max statin, use ezetimibe
 - if LDL still >1.8 & atherosclerotic CV disease, consider PCSK9 inhibitor (evolocumab)
- if on statin and TG still >1.5 & atherosclerotic CV disease or diabetes, consider icosapent ethyl 2g BID

+ non pharmacologic management
Smoking cessation
Diet and exercise prescriptions

Diabetes Management

- Must check: fasting glucose or HbA1c or 75gOGTT
- Target HbA1c ≤7%
 - if >7% despite standard OHA, consider SGLT-2 inhibitors or GLP-1 receptor agonists

GLP-1 Rc agonists reduce risk of nonfatal stroke in T2DM with either history of ASCVD or multiple CV risk factors.
(CCS 2022 Cardiorenal guidelines)

BONUS
Read on own

After the “Garden Variety” Stroke Secondary Prevention



Lifestyle Management

- Diet = high fruits/vegetable/low-fat dairy/fibre
 - protein from plant sources
 - low trans/saturated fat
 - low cholesterol (<200mg/day) = Mediterranean type diet
- Sodium <2000mg/day
- Exercise —> 150 min/ week moderate dynamic exercise [CCS]
(walking, jogging, cycling, swimming) 4-7days/week

AHA 2025 Stroke prevention guideline:

- In patients with periodontal disease, oral hygiene and regular dental care can lower stroke risk
- Address stroke risk factors in women with history of >2 miscarriages, small for gestational age, gestational diabetes, placental abruption or stillbirth.
- Migraine disorder, Endometriosis confers higher risk of stroke



BONUS
Read on own

After the “Garden Variety” Stroke Secondary Prevention



Lifestyle Management

- Weight = BMI 18.5 to 25
(waist <88cm for women, <102 for men)
- Alcohol = women <10dr/week, men <15dr/week
 - There is no safe amount of alcohol, in some cases counsel abstinence
- Smoking = discontinue
(via behavioural or pharmacological (NRT or Bupropion or Varenicline) approach - see Respiriology lecture for details)

After the “Garden Variety” Stroke Secondary Prevention



Post-stroke depression (PSD)

- 30% of stroke pts develop depressive symptoms
 - highest risk is within 1st year after stroke
- Risk factors for PSD
 - stroke severity
 - functional dependence
 - cognitive impairment
 - history of depression
- Screen in acute care, and at transition points
- Treatment:
 - mild depression, watchful waiting for 2-4 wks is ok
 - CBT or IPT, SSRI (if effective treat for minimum 6-12m)

After the “Garden Variety” Stroke Secondary Prevention



Post-stroke Vascular Cognitive Impairment (VCI)

- 1) All pts with stroke/TIA should be screened for VCI using a standardized tool (mini-cog, MOCA, MMSE, RUDAS)
- 2) Screening should be done in acute hospital, rehab hospital, and post-d/c
Note: in acute setting, delirium may confound assessment so defer if present
- 3) All pts with suspected VCI should have
 - 1) CBC, B12, TSH, Ca, lytes, Cr, ALT, lipid panel, HbA1c
 - 2) MRI Brain – lesions in left frontal, left temporal, left thalamic, right parietal are most likely to impar cognition
- 4) Management – HTN, DM, Lipids, Afib, Sleep D/O, Diet, Sodium, Exercise, Weight, Alcohol, Smoking, Mood

After the “Garden Variety” Stroke Secondary Prevention



Post-stroke Vascular Cognitive Impairment (VCI)

Management

- 1) HTN, DM, Lipids, Afib, Sleep D/O, Diet, Sodium, Exercise, Weight, Alcohol, Smoking, Mood
- 2) If Mixed (Alzheimers & Vascular) dementia, can use Acetylcholinesterase Inhibitors and/or memantine
- 3) Use ASA if you normally would have in stroke management
- 4) ASA for VCI who have covert infarcts on imaging WITHOUT history of stroke has not been defined. Can consider it, but benefit unclear.

Stroke

1. Hyperacute management of ISCHEMIC stroke (in the ER or ICU)
 - Thrombolysis, Antiplatelets, BP Targets, Seizure management and more...
 - Hyperacute management of HEMORRHAGIC stroke
2. Secondary Workup of stroke or TIA
3. Management by Mechanism and Secondary Stroke Prevention
 - Cardio-Embolic: AFIB, PFO, ESUS
 - Athero-Embolic: Carotid stenosis, Dissection
 - Lacunar and other “Garden variety” stroke secondary prevention
4. **Stroke in Special Populations: Young or Pregnant, Cerebral Venous Thrombosis (CVT = **NEW in 2024!**), Cancer associated Stroke**
5. Stroke Syndromes to Recognize for your exam



Stroke in the Young (age <55)

Large and broad differential but...

- Common things are still common
 - Atherosclerosis of large arteries (8%), small vessel disease (14%)
 - HTN, Smoking, Dyslipidemia, DM
 - Cardioembolic (20%)
 - Dissection (15%)
 - Others less frequently but to consider in this age group... (next slide)

Because at least 10 of you ask us every year:

“Stroke in a young person, routine w/u negative - what one test to order next?”

Cardioembolic >>> APLA/thrombophilia so if you have to pick on MCQ, **pick 14 d holter or TTE w bubbles over APLA!**

Stroke in the Young

- DDx (think usual stroke causes + VITAMIN-C)
 - **Cardioembolic** – AFIB, PFO, Endocarditis, structural heart disease
 - Large Artery - **dissection**, Moya-Moya, Fibromuscular dysplasia
 - Small Vessel - RCVS Reversible Cerebral vasoconstriction syndrome, Susac Syndrome

Vascular - cerebral venous thrombosis, dissection etc

Infection - viral (**VZV**), bacterial (**syphilis, TB**), Fungal (aspergillus), parasitic (neurocysticercosis)

Toxins – Cocaine/ OCP/pregnancy

Autoimmune - Inflammatory/Vasculitis - primary CNS vasculitis, secondary to systemic vasculitis (GCA, takayasu, PAN, GPA, eGPA, cryoglobulinemia, SLE, sjogrens, RA, Behcets)

Metabolic / Hypercoagulable - APLA, Polycythemia Vera, essential thrombocytosis, TTP, preeclampsia

Idiopathic

Neoplastic –intravascular lymphoma, other cancers, chemotherapy (L-asparaginase), radiation

Congenital - CADASIL, CARASIL, HHT, Sickle Cell, Fabry's, MELAS

Stroke in the Young

Work-up

- Usual stroke w/u = MRI, CTA, TTE with bubble, 14d Holter, LDL, HbA1c
- Bloodwork:
 - CBC, INR, PTT
 - Inflammatory: ESR, CRP, RF, ANA/ENA, ANCA, C3/4
 - Infectious: viral serology, bacterial/fungal cultures
 - Hypercoagulable studies: APLA/venous hypercoag testing now & in 3 months
 - Mitochondrial: lactate, pyruvate, gene testing
- MRV, TEE, LP, Malignancy screen

Cancer and Stroke

- In pts with active malignancy & stroke in whom a cancer-associated hypercoagulable state may be the cause, anticoagulation could be considered over antiplatelet
- When anticoagulation is used, LMWH is preferred
Role of DOACs is unknown but under study and may be reasonable after consideration of patient preference

This will not be an MCQ on exam. Workup of extensive stroke in patient with malignancy would be more seen on applied / oral exam.
Think MARANTIC endocarditis if multiple embolic appearing strokes.

Pregnancy and Stroke



- Not high yield for your exam
- There are consensus guidelines (strokebestpractices.ca 2018 Guidelines - lacking RCTs in this population) summarized in following bonus slides
- Case by case management!

Stroke in Pregnancy (1)



Pre-pregnancy counselling for women who have had a history of stroke

- Counsel re risk of recurrent stroke in pregnancy, consider referral to specialized team
- Stroke risk in pregnant women 3X that of non pregnant women in general
- Discuss contraceptive options
 - Avoid combined oral contraceptives/estrogens in patients with history of stroke
 - Progestin only options considered safe (balanced with risk of unplanned pregnancy)
- Lifestyle modifications as in all patients, review re: etiology of stroke Review of medications (ACE, statin, antiplatelets)
- Change to low dose ASA 81mg preferred anti platelet, suggest change PRIOR to pregnancy [insufficient evidence on safety of others]
- Warfarin avoided 1st trimester (—>LMWH). See OB Med lecture for details
- Case by case consideration of systemic anticoagulation (ex APLA)

Stroke in Pregnancy (2)



Secondary Stroke Prevention during pregnancy

- Anti platelet (ASA!) and Anticoag as in (1)
All women with history of stroke: BP <140/90 (see OB Med lecture for choice antihypertensives in pregnancy)
- Do not routinely measure lipids and adjust therapy in pregnancy as they will raise in pregnancy
- Data on statins evolving – see OB Med lecture
- Diabetic control as per OB Med

Stroke in Pregnancy (3)



Management of Acute Stroke in pregnancy

- Etiology:
 - Pregnancy Related: HELLP, Preeclampsia, RCVS
 - Patient Related: Cardioembolic (mech valve, PFO, valvular Afib), APLA / thrombophilia
 - Other see “Stroke in the young”: Endocarditis, Drugs (cocaine), Dissection, etc
- Preeclampsia represent the majority (treat as per preeclampsia scenario)
 - **BP TARGET** for severe preeclampsia (incl. stroke) —> **SBP <160/110**
 - Thrombolysis/EVT Candidate?
 - Pregnancy was an exclusion criteria in all trials = case by case (if delivery EMERGENT e.g. preeclampsia, thrombolysis is off the table!), “expert consultation recommended”
 - tPA is a large molecule and doesn’t cross placenta
 - EVT could be considered case by case
- ICH in pregnancy – BP control and imaging to rule out aneurysm
- Cervical artery dissection → Not a contraindication to vaginal delivery
- Cerebral Sinus Venous Thrombosis
 - Transfer to centre with expertise (neurosurg)
 - Rx LMWH therapeutic dose (or IVuFH if baby still in and planning delivery)

Cerebral Venous Sinus Thrombosis

NEW [Strokebestpractices.ca](https://strokebestpractices.ca) guideline 2024

Symptoms	Demographics
Headache >80%	Female >70% *younger
Seizure >25%	OCP / Pregnancy
Focal Neuro Deficits >30%	Hereditary thrombophilia
Depressed LOC >20%	Obesity
Vision change, diplopia, cranial nerve changes > 10%	APLAS Others: malignancy, iatrogenic (eg post neurosx), connective tissue diseases

- WORKUP Acutely:
 - Imaging: CT/CTV or MRI/MRV (non-contrast CT head & time-of-flight MRV is not sensitive/specific enough)
 - Labs: CBC, lytes, Cr, INR, aPTT, bHCG, Glc, ALT, TSH
 - D-dimer not helpful to rule in/out
 - LP not recommended
- Workup in followup
 - APLAS
 - Testing for inherited thrombophilia controversial

Cerebral Venous Sinus Thrombosis

NEW [Strokebestpractices.ca](https://strokebestpractices.ca) guideline 2024

- ACUTE Management
 - ANTICOAGULATION
 - Indicated even if ICH on initial scan (seen in 30-40% of initial CT w CVT)
 - LMWH preferred over IV uFH unless planning surgery (eg hemicraniectomy for ICP)
 - Insufficient evidence to recommend DOAC first line
 - ICH / SAH is not a contraindication to anticoag in this scenario (case by case)
 - Seizures
 - Acute seizure (within 7d of presentation) should be treated with AED (often for 3 months)
 - Late onset after 7d may require long term (>3months) AEDs
 - EEG if intermittent dec LOC to rule out non convulsive status
 - ICP
 - Ophthalmologic exam for papilledema
 - Acetazolamide for papilledema
 - Failing medical therapy – neurosurgery consult urgently
 - Endovascular therapy is not standard of care and controversial

strokebestpractices.ca

Cerebral Venous Sinus Thrombosis

NEW [Strokebestpractices.ca](https://strokebestpractices.ca) guideline 2024

- POST - acute Management
 - ANTICOAGULATION
 - Once clinically stable, should transition to DOAC or Warfarin (target INR 2-3)
 - Use Warfarin if APLA syndrome discovered
 - DURATION of anticoagulation
 - CVT with major transient risk factor (e.g. OCP use) = 3-6 months
 - 1st CVT with no prior VTE or risk factor = 6-12 months
 - CVT with major chronic risk factor (e.g. cancer, prior VTE, hypercoag dx) = indefinite

Stroke

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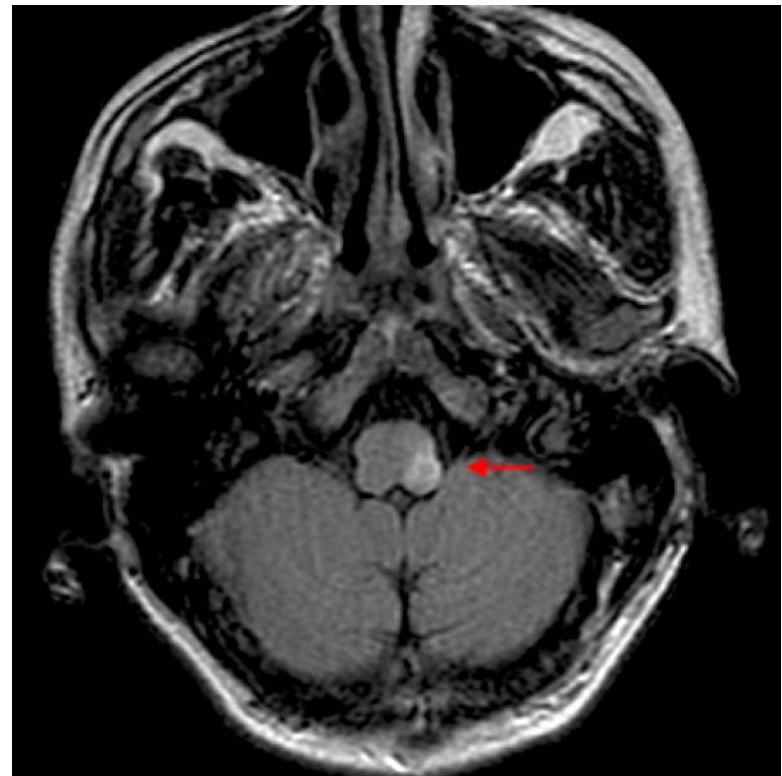
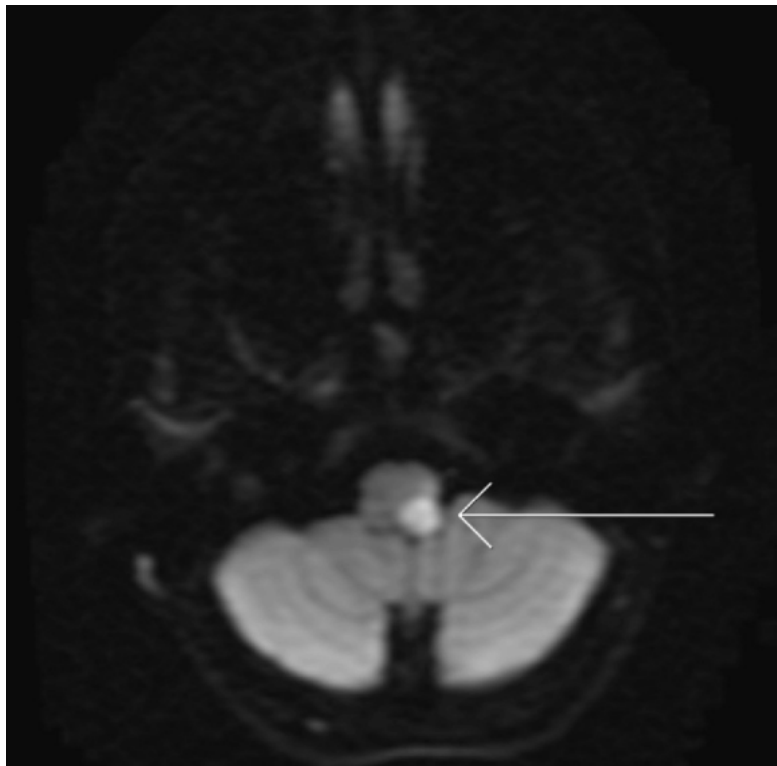
MCQ #2 2026

63M presents acutely with vertigo, ataxia, right ptosis, left arm and leg numbness and hiccups. What is the most likely etiology?

- a) left paramedian pontine infarct
- b) multiple sclerosis
- c) right lateral medullary infarct
- d) right midbrain infarct
- e) drug or alcohol intoxication

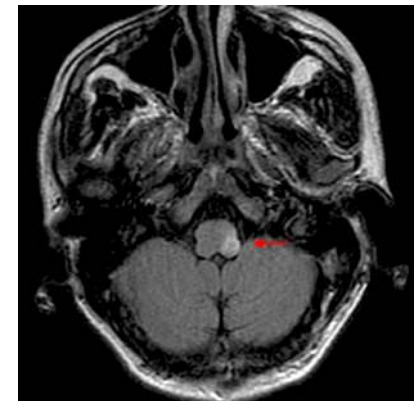
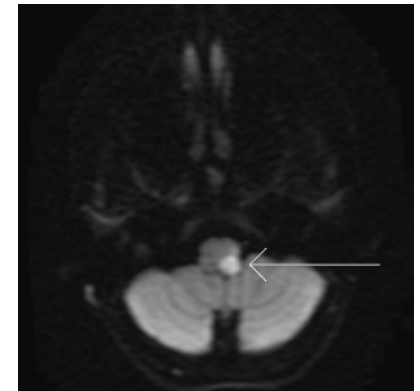
Stroke Syndromes

Examine this patient...



Stroke Syndromes

Localization	Eponym	Vessel	Sx/Sx
Lateral Medullary Syndrome	Wallenberg	PICA or Vertebral	- ipsilateral ataxia - ipsilateral Horner's (ptosis, miosis, anhidrosis) - ipsilateral face loss of pain/temp sense - contralateral body loss of pain/temp sense - vertigo & nystagmus - dysphagia & hoarseness - hiccups

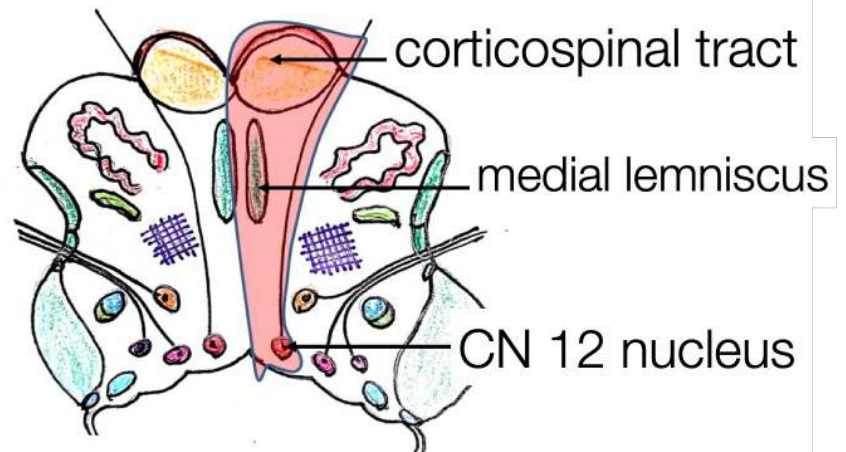
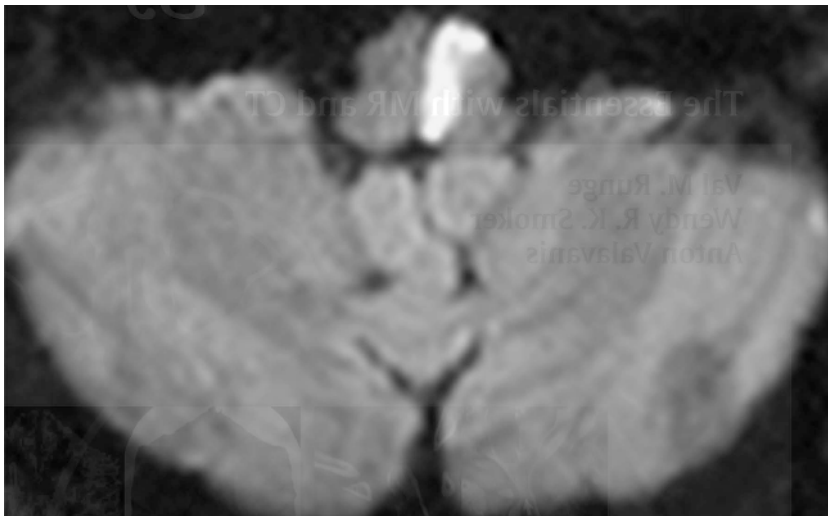


Stroke Syndromes

Exam	Findings in Lateral Medullary Syndrome
Vitals	HR ?afib BP ?elevated
CN	Ipsilateral Horner's (ptosis, miosis, anhidrosis) Ipsilateral loss of face pain/temp sense (& reduced corneal reflex) Ipsilateral absent/reduced gag reflex Nystagmus (fast phase beating away from lesion) Hoarse voice
Motor	Normal strength Normal reflexes
Sensory	Contralateral loss of body pain/temp sense
Coordination	Ipsilateral dysmetria with Finger-to-Nose & Heel-to-Shin Ipsilateral dysdiadochokinesia with rapid alternating movements
Gait	Ataxia (wide-based, staggering gait)
Other	Hiccups

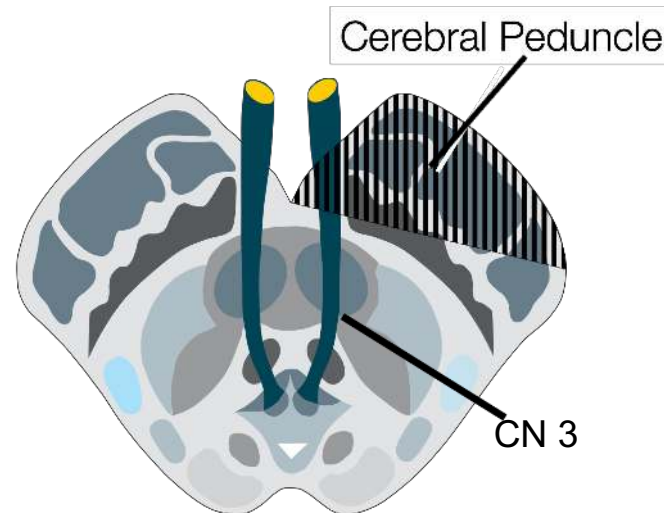
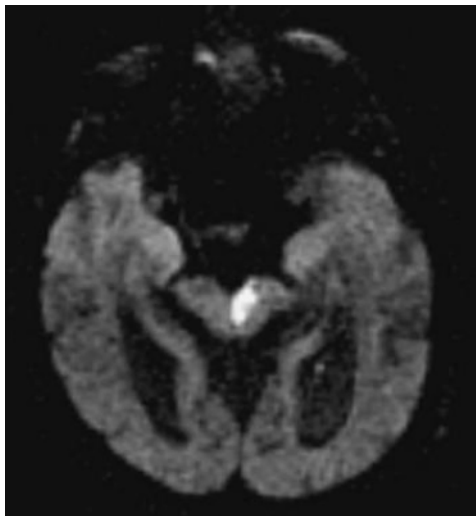
Stroke Syndromes

Localization	Eponym	Vessel	Sx/Sx
Medial Medullary Syndrome	n/a	Anterior Spinal Artery	- ipsilateral tongue weakness - contralateral arm, leg weakness - contralateral body vibration, proprio loss



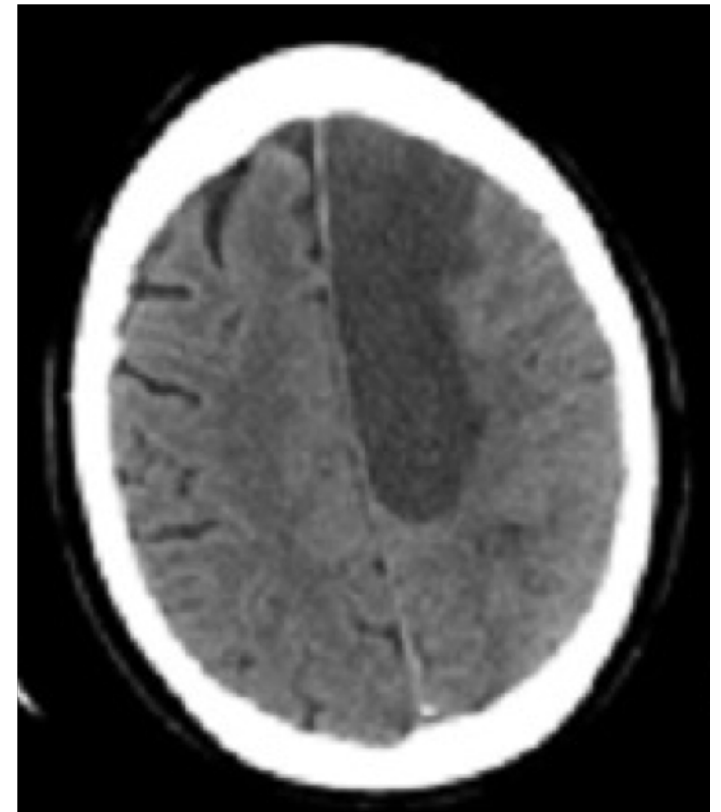
Stroke Syndromes

Localization	Eponym	Vessel	Sx/Sx
Midbrain	Weber	PCA	- ipsilateral CN 3 palsy (ptosis, mydriasis, diplopia (down & out)) - contralateral hemiplegia (face, arm, leg)



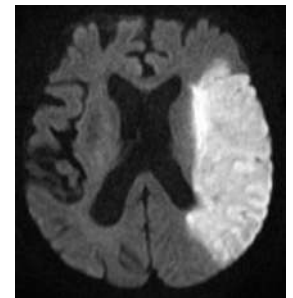
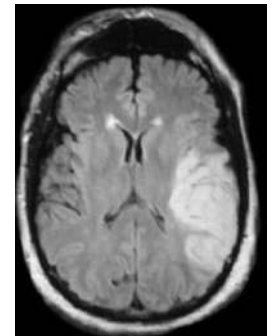
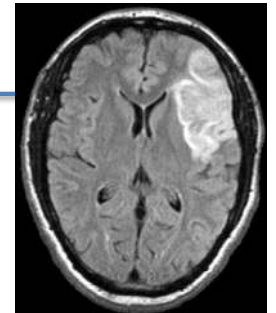
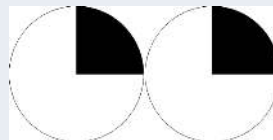
Stroke Syndromes

Vessel	Sx/Sx
ACA	- Contralateral leg weakness - Contralateral leg numbness - Contralateral grasp reflex (and other frontal signs)

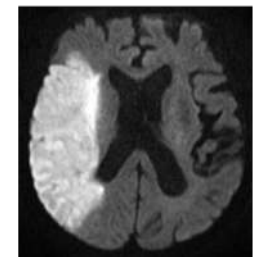
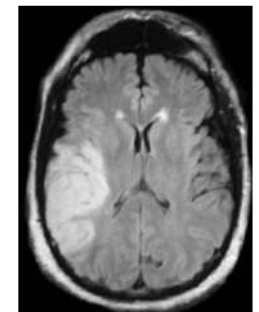
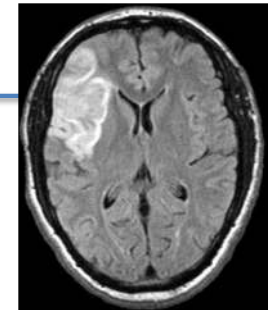


Stroke Syndromes

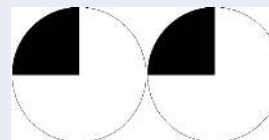
Vessel	Sx/Sx
LEFT MCA - superior branch	Broca's (expressive/non-fluent) Aphasia Right weakness face & arm > leg Gaze deviation to left
LEFT MCA - inferior branch	Wernicke's (receptive/fluent) Aphasia Right cortical sensory loss Right "pie-in-the-sky" = right homonymous superior quadrantonopsia
LEFT MCA - main branch (M1)	all of the above global aphasia



Stroke Syndromes

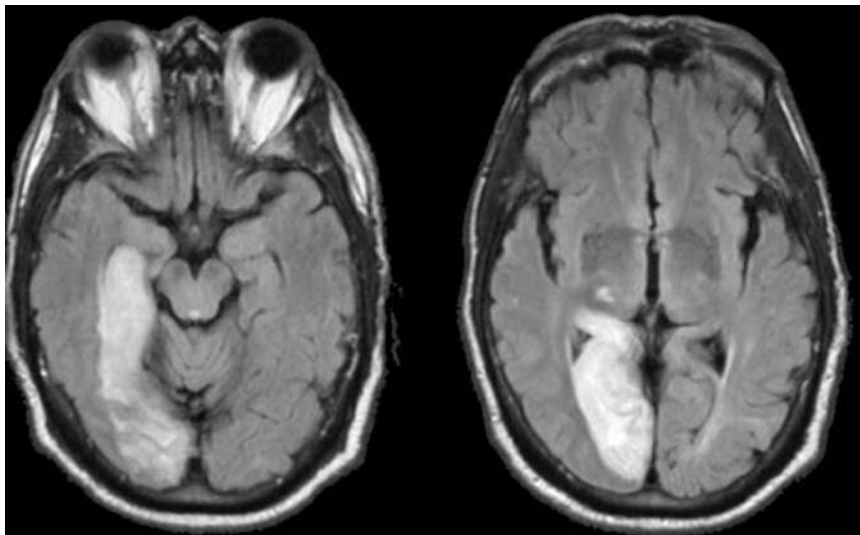


Vessel	Sx/Sx
RIGHT MCA - superior branch	Left weakness face & arm > leg Gaze deviation to right
RIGHT MCA - inferior branch	Left cortical sensory loss Left hemineglect Left “pie-in-the-sky” = left homonymous superior quadrantonopsia
RIGHT MCA - main branch (M1)	all of the above

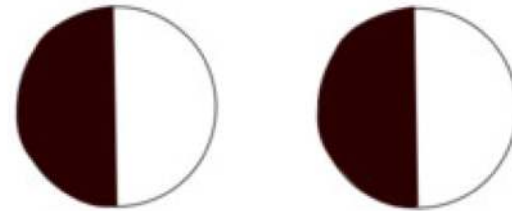


Stroke Syndromes

Vessel	Sx/Sx
PCA	contralateral homonymous hemianopia



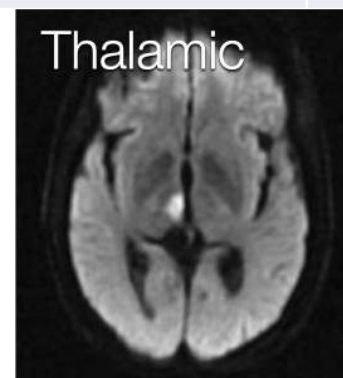
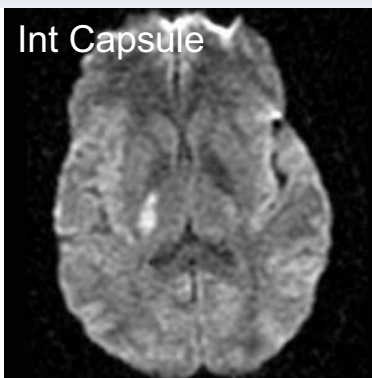
R PCA stroke



Left homonymous
hemianopia

Stroke Syndromes

Lacunar Syndrome	Possible Localizations	Artery	Sx/Sx
Pure Motor	Posterior Limb Internal Capsule Corona Radiata Midbrain - cerebral peduncle Ventral Pons	Anterior Choroidal Small MCA branch Small PCA branch Basilar	contralateral face, arm, & leg weak
Pure Sensory	Thalamus	Thalamoperforators from PCA or MCA	contralateral face, arm, leg sensory sx



MCQ #2 2026

63M presents acutely with vertigo, ataxia, right ptosis, left arm and leg numbness and hiccups. What is the most likely etiology?

- a) left paramedian pontine infarct
- b) multiple sclerosis
- c) right lateral medullary infarct**
- d) right midbrain infarct
- e) drug or alcohol intoxication

Answer: C - Several, though not all, features of the lateral medullary syndrome

Multiple sclerosis

NEW diagnostic criteria follow!

Beware old IMR / exam MCQs
on diagnosis of MS are based
on OLD diagnostic criteria!



Multiple sclerosis

Types of MS

RRMS (relapsing remitting MS - 90%)



50% conversion in 10y

90% conversion in 25y

SPMS (secondary progressive MS)

PPMS (primary progressive MS - 10%)



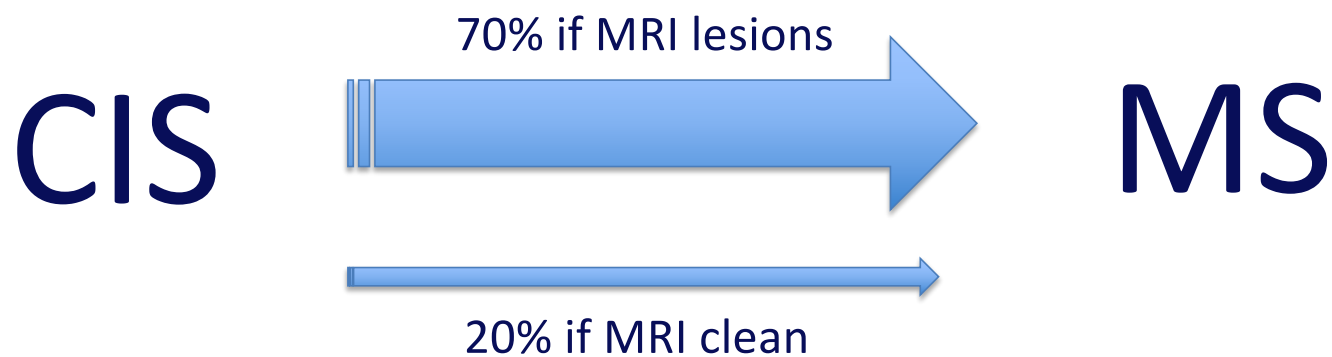
Multiple sclerosis

Epidemiology of RRMS

- Most common demyelinating disease of CNS
- Affects 0.26% of adults, higher in northern Europeans
- 3% risk in 1st degree relative
- 3F:1M
- Mean age of onset: 30 yo
- Rare onset <15yo or >60yo (consider another disease)

Clinically Isolated Syndrome (CIS) = First demyelinating attack

- monophasic clinical episode
- acute or subacute onset **lasting >24h**
- symptoms and objective findings typical of a focal/multifocal demyelinating CNS event
- In absence of fever/infection
- E.g. first episode of optic neuritis



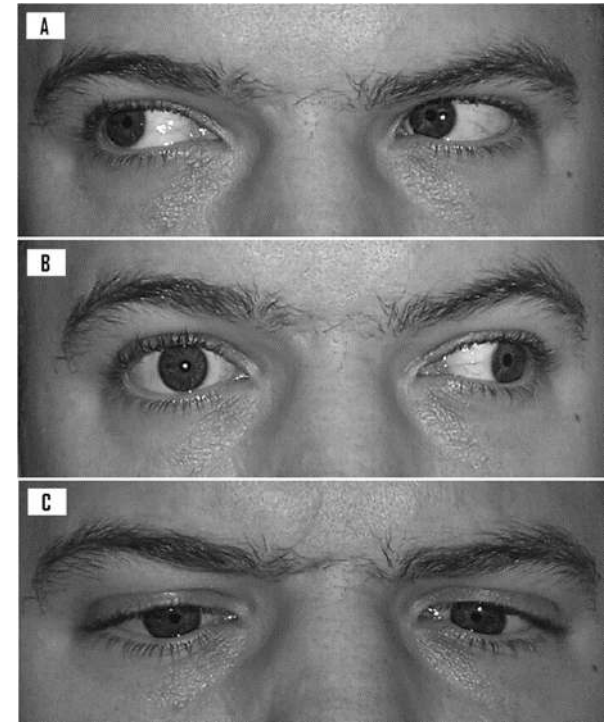
Multiple sclerosis: Typical syndromes

Optic neuritis: monocular vision & colour loss, painful eye movements, RAPD, +/- disc edema

Brainstem/Cerebellar syndrome: INO, diplopia, dysarthria, ataxia, gaze-evoked nystagmus, vertigo, facial numbness

Partial transverse myelitis: sensory loss, asymmetric limb weakness, urge incontinence, erectile dysfunction, Lhermitte (neck flexion → spinal electric shock sensation)

Uhthoff phenomenon (heat exacerbation)



Right INO: impaired adduction on leftward gaze

Differential Diagnosis of MS

Demyelinating:	ADEM, NMO*, MOG Antibody Disorder*
Inflammatory:	Sarcoidosis, SLE, Sjögren's, GPA, Behçet's
Infectious:	post-infectious demyelination, HIV, PML, HTLV, syphilis, Lyme, Bartonella, TB
Metabolic:	B12 or Cu deficiency
Neoplastic:	lymphoma
Psychiatric:	somatization
Vascular:	vasculitis, vascular malformation, migraine

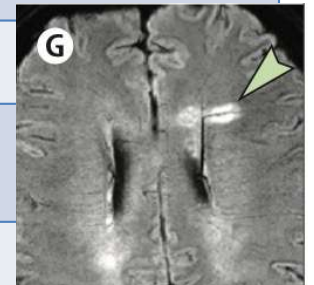
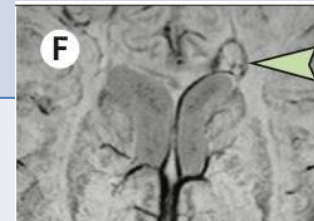
***RARE MS mimics – not all optic neuritis is MS.**

NMO = Neuromyelitis optica – Antibody to Aquaporin4 Channel – any combo of optic neuritis, myelitis.
MOGAD = Anti myelin oligodendrocyte antibody disorder, can cause Optic Neuritis, Transverse Myelitis or ADEM or unilateral encephalitis
Clinical overlap – ddx largely by MRI findings hence well beyond scope of GIM exam.
Rx: IV steroids acutely for treatment.

MS: a few definitions...



CIS	Clinically isolated syndrome = 1 clinical attack
RIS	Radiographic isolated syndrome = incidental discovery of CNS lesion typical for MS on MRI, in absence of MS symptoms/signs
DIS	Dissemination in space (DIS) = ≥ 2 clinical attacks or ≥ 2 of 5 CNS Sites 5 sites: optic nerve , cortical/ juxtacortical, periventricular, infratentorial, or spinal cord
DIT	Dissemination in time (DIT) = ≥ 2 clinical attacks or 1 MRI has Gad-enhancing and non-enhancing lesions or 2 separate MRIs show new T2 or Gad lesion on the follow-up scan
OCB	Oligoclonal Bands (OCB)– an indicator of intrathecal immunoglobulin synthesis
kFLC	Kappa Free light chain index is an alternative to OCBs – largely unavailable in Canada!
CVS	Central Vein Sign – ≥ 6 venocentric demyelinating lesions on SWI or T2* (G)
PRL	Paramagnetic Rim Lesion - ≥ 1 of SWI or T2* hypointense rim around an MS lesion (F)
Optic nerve	Optic nerve involvement can count as site if confirmed with MRI, Optical CT or Visual evoked potentials



2024 McDonald Criteria for MS



Pt presents with signs/symptoms of MS (1st clinical attack):

If MRI shows lesions in **≥2 CNS sites**

OR

≥12months progressive symptoms AND ≥2 spinal lesions

THEN

MS is the diagnosis if 1 of:

Moving to
biomarker
based
diagnosis

- { Oligoclonal banding or kappa free light chain in CSF+
- CVS Positivity +
- Dissemination in time +
- Lesions in 4 or all 5 CNS sites

2024 McDonald Criteria for MS



Pt presents with signs/symptoms of MS (1st clinical attack):

If MRI shows lesions in **1 CNS site**
including pts with ≥ 12 months progressive symptoms

THEN

MS is the diagnosis if 1 or more of:

Oligoclonal banding or kappa free light chain in CSF+ AND CVS+

Oligoclonal banding or kappa free light chain in CSF+ AND PRL+

Dissemination in time + and CSF + or DIT + PRL +

Beware old IMR / exam MCQs
on diagnosis of MS are based
on OLD diagnostic criteria!

2024 McDonald Criteria for MS

Said another way,

Diagnosis of MS can now be made in the following:

DIS + DIT (unchanged from 2017)

DIS + OCB+ (unchanged from 2017)

CIS + 4 CNS locations (note changes: *DIT is no longer necessary*, **optic nerve now one of the CNS locations**)

CIS + 1 CNS location + DIT **or** OCB+ and 1 of CVS+ or PRL+

2024 McDonald Criteria for MS

Pt presents with incidental imaging findings suggestive of demyelination =
Radiographic isolated syndrome (RIS)

If MRI shows lesions in 2 CNS sites

THEN

MS is the diagnosis if any 1 of:

OCB+

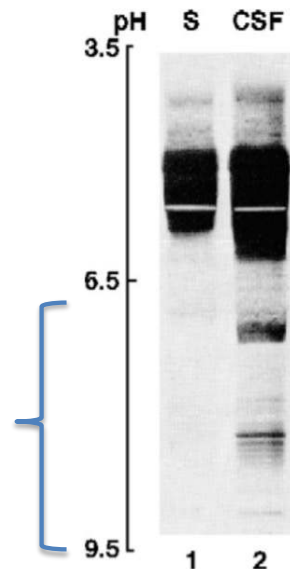
CVS+

DIT

Investigations

- MRI brain & cervical-thoracic spine with gadolinium
- ± LP for oligoclonal bands, IgG index

Although in NEW diagnostic criteria, kappa free light chain index is not available in Canada.



- CBC, Cr, liver enzymes, SPEP
- B12, TSH
- ± NMO/MOG antibody
- ± ANA, ESR, CRP, ENA, ACE, ANCA
- ± VDRL, Lyme, HIV, HTLV
- ± visual evoked potentials

ENA = extractable nuclear antigen antibodies
ACE = angiotensin converting enzyme
NMO = neuromyelitis optica, also known as anti-aquaporin 4 antibody
MOG = myelin oligodendrocyte protein
HTLV = human T-lymphotropic virus

Rose JD, 2000

Treating MS attacks

- If functionally disabling, high dose corticosteroid can **speed** recovery
 - Used in optic neuritis, motor weakness; not mild sensory symptoms
- Does **not** alter degree of recovery
- Does **not** reduce future attack risk

Methylprednisolone 1000 mg IV daily for 3-7 days ± taper

Alternative: PO steroid appears equally effective (COPOUSEP trial), e.g. prednisone 1250 mg for 3-7 days ± taper

Plasmapheresis

If poor steroid response

Disease-modifying therapy in RRMS

- ✓ Decrease RRMS relapse rate
 - ✓ Decelerate MRI brain lesion accumulation
 - ✓ Lower risk of conversion to SPMS

 - Injections: beta-interferon, glatiramer acetate
 - Oral agents: dimethyl fumarate, teriflunomide, fingolimod, siponimod, cladribine
 - Infusions: natalizumab, alemtuzumab, ocrelizumab
 - Hematopoietic stem cell transplant
- Lifestyle
- Physical activity
 - Smoking cessation
 - Vitamin D 4000 IU daily

Movement Disorders



MCQ #3 – 2026

A 68yo patient, from a nursing home, with a history of depression presents to hospital with symmetrical parkinsonism including rigidity and bradykinesia 3 months after a laparoscopic cholecystectomy. What is the most likely culprit?

- a) escitalopram
- b) venlafaxine
- c) clozapine
- d) metoclopramide
- e) domperidone

Approach to Tremor

Diagnosis?

- a) Parkinson Disease
- b) Essential Tremor
- c) Dystonic Tremor
- d) Wilson's Disease
- e) Psychogenic



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Approach to Tremor

Diagnosis?

- a) Parkinson Disease
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Approach to Tremor

Types of Tremor and Common Causes

1. Rest

Idiopathic Parkinson Disease

Other Parkinsonism (Drug-induced, LBD, VascPD)

2. Action

1. Postural

Enhanced Physiologic Tremor

Essential Tremor

Dystonic Tremor

2. Kinetic/Intention

Cerebellar Disease (MS, stroke etc)

Lithium

Note: Psychogenic Tremor can present any type

Approach to Tremor

Examination for Tremor

1. Check with arms at rest
—> then add a cognitively distracting task e.g. Serial 7's
2. Check with arms outstretched
3. Check with finger-to-nose testing
4. Check with water pouring task
5. Look at the company the tremor keeps
 - i.e. masked face, hypophonia, rigidity = PD
 - i.e. +ve family history, improves with alcohol = ET
 - i.e. head tilt & head tremor = Dystonic tremor

Approach to Tremor

	PD Tremor	ET Tremor
Frequency	4-6Hz	8-10Hz
Symmetry	asymmetrical	symmetrical
Position	rest	action/posture
Family History	rare	common (?50%)
Response to LDopa	good	none
Response to Alcohol	none	good (~50%)
Level of impairment	minimal	significant
Assoc writing	micrographia	messy/tremulous

Approach to Tremor

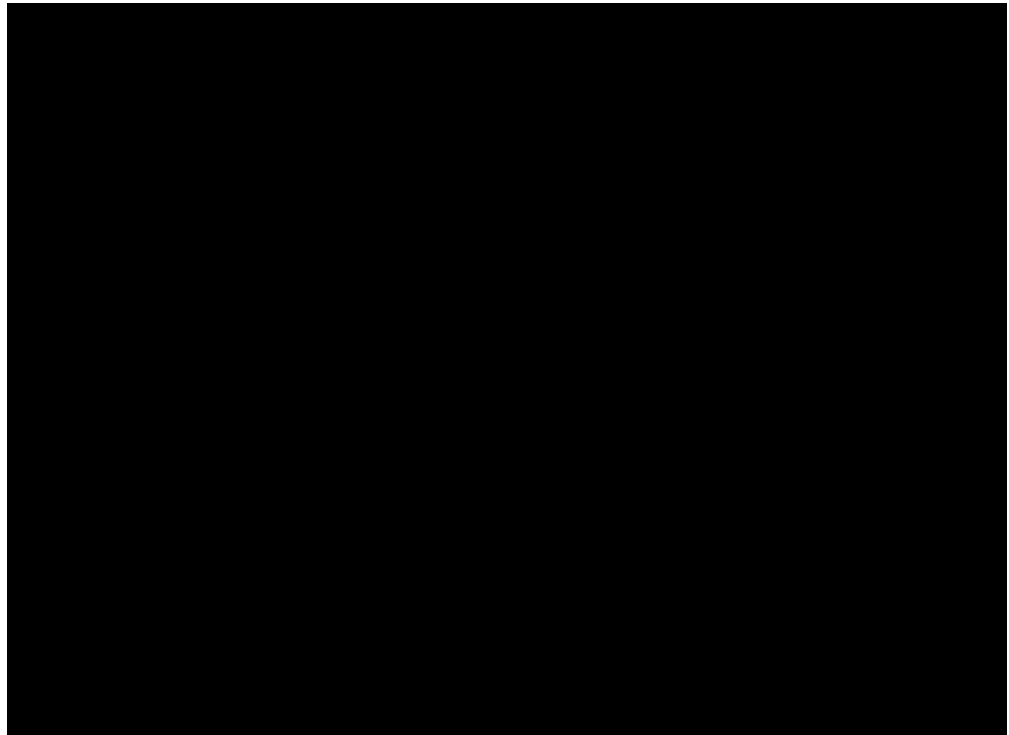
Enhanced Physiologic Tremor

- high frequency
- small amplitude
- symmetrical
- postural, not rest
- “enhanced” by caffeine, anxiety, stress, hyperthyroidism, drugs (cocaine, amphetamines), withdrawal from alcohol

Approach to Tremor

Dystonic Tremor

- seen in patients with dystonia, when they “fight” their dystonic posture
- can occur in head & hands
- asymmetrical & irregular
- postural, not rest



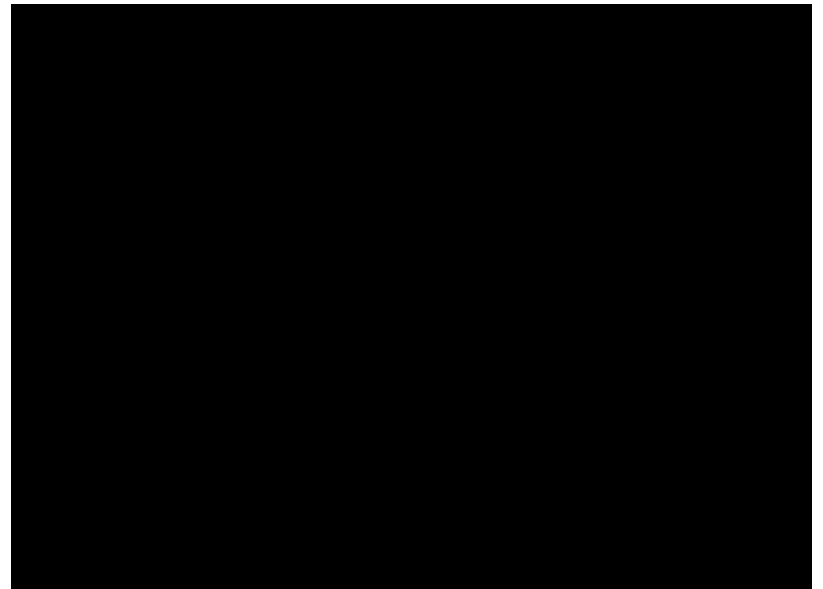
Approach to Tremor

Cerebellar Tremor

- Intention tremor (can be posture as well)
- Can be associated with other cerebellar signs such as dysdiadochokinesia, gait ataxia
- If asymmetrical, sudden onset,

older patient with vascular risk factors

- -> think stroke
 - If young -> think MS or Wilson's



Approach to Tremor

Psychogenic Tremor

- Rest or posture or intention or mix of all three
- Irregular
- Variable frequency and amplitude
- Distractible
- Entrainable

Approach to Tremor

Diagnosis?

- a) Parkinson Disease
- b) Essential Tremor
- c) Dystonic Tremor
- d) Wilson's Disease
- e) Psychogenic



Approach to Tremor

Diagnosis?

- a) Parkinson Disease
- b) Essential Tremor
- c) Dystonic Tremor
- d) Wilson's Disease
- e) Psychogenic



Parkinsonism

Parkinsonism ≠ Idiopathic Parkinson Disease

Parkinsonism = “TRAP”

1. Akinesia/Bradykinesia

- slowness but most importantly, fatiguing + decrement in size of repetitive movement

2. Rigidity

- resistance when moving the limb, same throughout mvmt in flex/ext

3. Tremor

- rest tremor, 4-6Hz, usually of thumb, re-emergence of tremor

4. Postural Instability

- festinant gait = short shuffling steps, reduced arm swing, sometimes camptocormic, later freezing (in narrow spaces or doorways)

Parkinsonism

Differential Diagnosis of Parkinsonism

- Idiopathic Parkinson disease
- Parkinson plus disorders
 - LBD, PSP, MSA, CBD
- Vascular parkinsonism (stroke)
- Drug-induced parkinsonism
 - Antipsychotics
 - Metaclopramide
- Genetic
 - Wilson's disease
- Toxins
 - Manganese

Idiopathic Parkinson Disease

Parkinsonism ≠ Idiopathic Parkinson Disease

To Diagnose Idiopathic Parkinson Disease =

1. must have "Parkinsonism"

Akinesia/Bradykinesia

and one of

Rigidity *or* Tremor

2. absence of absolute exclusion criteria

3. two supportive criteria

4. no red flags

Idiopathic Parkinson Disease

Absolute **Exclusion** Criteria

1. Cerebellar abnormalities (think MSA)
2. Downward vertical gaze palsy/slowing of vertical saccades (think PSP)
3. Diagnosis of frontotemporal dementia (FTD)
4. Exclusively lower extremity parkinsonism x 3yrs (think vascular PD)
5. Treatment with a dopamine receptor blocker (think drug-induced PD)
6. Absence of observable response to levodopa
7. Cortical sensory loss, apraxia, or aphasia (think CBD)
8. Normal functional neuroimaging of the dopaminergic system
9. Documentation of an alternative diagnosis that can cause parkinsonism

Idiopathic Parkinson Disease

Supportive Criteria (need at least 2)

1. Clear response to L-Dopa
2. Presence of L-Dopa induced dyskinesia
3. Rest tremor of a limb
4. Documented olfactory loss or cardiac sympathetic denervation on MIBG scintigraphy

Idiopathic Parkinson Disease



Red Flags that this is not idiopathic PD

1. Rapid gait impairment requiring wheelchair within 5 y of onset
2. Complete absence of progression of motor symptoms
3. Early bulbar dysfunction
4. Stridor
5. Severe autonomic failure in first 5yrs of onset
6. Recurrent falls within 3 years of onset
7. Anterocollis
 - Anterior neck flexion $>45^\circ$ due to dystonia of neck flexors - beware MSA
8. Absence of common non-motor symptoms despite 5 yrs of disease
9. Pyramidal signs [UMN findings]
10. Bilateral symmetric Parkinsonism

Idiopathic Parkinson Disease

The Typical Idiopathic PD

1. Preceding Illness

- anosmia
- REM-Behaviour Sleep Disorder
- constipation

2. Early Illness

- asymmetric tremor
- asymmetric bradykinesia
- asymmetric rigidity

3. Late Illness

- postural instability/falls
- non-motor manifestations
 - dementia
 - visual hallucinations (can be from meds as well)
 - autonomic dysfunction = postural hypotension (can be from meds as well)

Treatment of Parkinson Disease

Medication	Indication	Mechanism	Side Effects
Levodopa/ Carbidopa	PD motor sx	converts to dopamine in CNS	Non-motor: n/v, orthostatic hypotension, sedation, confusion, hallucinations Motor: dyskinesias Special: dopa dysregulation (addiction)
Pramipexole	PD motor sx	dopamine agonist	Same as L-dopa but more orthostatic hypotension, sedation (sleep attacks), hallucinations + Impulse control disorder (gambling, sexuality, shopping, eating), leg swelling
Trihexyphenidyl	tremor	anticholinergic	Anticholinergic (confusion, dry eyes, blurry vision, dry mouth, constipation, urinary retention) + memory difficulty

Treatment of Parkinson Disease

Medication	Indication	Mechanism	Side Effects
Selegeline	PD motor sx	MAO-B inhibitor	Above 10mg, HTN crisis if eat tyramine rich foods (aged cheese/sausage/red wine) theoretical risk of serotonin syndrome alerting, insomnia, hallucinations, dry mouth, hypotension
Entacapone	prolong action of L-dopa	COMT inhibitor	Same SE as increasing sinemet, but also orange urine
Amantadine	dyskinesias	multiple mech	Anticholinergic (confusion, dry eyes, blurry vision, dry mouth, constipation, urinary retention), insomnia, levido reticularis, hallucinations Contraindicated in pts with sz
Domperidone	orthostatic hypotension	dopamine antagonist in gut	Prolongs QT interval (if above 10mgTID)

Effect vs Side Effect in Treatment of PD

	Levodopa	Dopamine Agonists (pramipexole, ropinorole)	MAO-B Inhib (selegiline, rasagiline)
Motor Sx Improvement	+++	++	+
Motor Complications	+++	++	+
Specific Adverse Events (ICD, somnolence, hallucinations)	++	+++	+

A question we get asked every year:

Young PD pt (<60yo), afraid of developing dyskinesias & has...

a) mild PD symptoms = choose MAO-B Inhibitor

or

b) moderate/severe PD symptoms = choose Levodopa or DA agonist (no right answer between the two)

2019 Canadian Guideline for Parkinson Disease

Diagnosis

- suspect PD if tremor, stiffness, slowness, balance symptoms
- CT/MRI should not be routinely used to diagnose PD

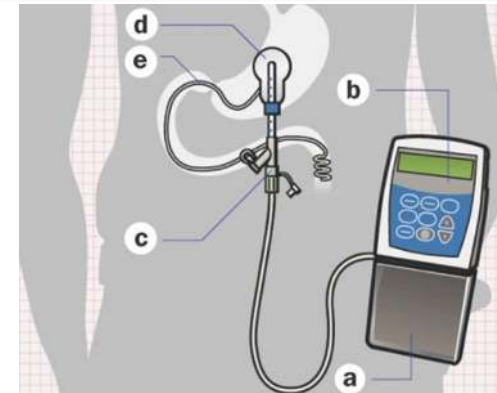
Pharmacologic Treatment:

- No therapy is truly “neuroprotective”
 - Levodopa is most effective med and can be used early
- Impulse control disorders can develop on any dopaminergic therapy, at any stage of disease, but most commonly with dopamine agonists
- Intrajejunal gel infusion (continuous duodopa)

Surgical Treatment: Deep brain stimulation

Non Pharmacologic Treatment:

- Rehab therapists (PT/OT/SLP) can help at any stage of disease
- Regular exercise program started early has proven benefit



FYI only - advanced therapies for PD
- refer to mvmt disorders clinic



2019 Canadian Guideline for Parkinson Disease

Non Motor Features

- botulinum toxin A helps sialorrhea
- orthostatic hypotension can be treated with midodrine, fludrocortisone, and domperidone
- REM behaviour sleep disorder can pre-date the diagnosis of PD

Palliative Care

- palliative care needs should be considered at all stages of disease
- if pt asks, option of medical assistance in dying should be discussed

Drug-Induced Parkinsonism



Scenario

- Patient on antipsychotics (typicals more than atypicals) or metoclopramide post operatively that wasn't stopped

Clinical Features

- Symmetrical parkinsonism (rigidity/bradykinesia)
- Tremor is usually postural, more than rest
- Commonly occurs within weeks or months of onset of drug and is reversible after drug is stopped over months

Meds that induce/exacerbate parkinsonism



Class	Example	Notes
Typical/First Generation Antipsychotics	Haloperidol Chlorpromazine Loxapine	
Atypical/Second Generation Antipsychotics	Risperidone Olanzapine Ziprasidone Palliperidone Aripiprazole	Quetiapine and Clozapine are safer alternatives
Antiemetics	Metoclopramide Prochlorperazine	Domperidone is a safer alternative as it doesn't cross BBB
Mood Stabilizers	Valproic Acid Lithium	
SSRI	Citalopram/Escitalopram Sertraline Fluoxetine	Case reports/case series

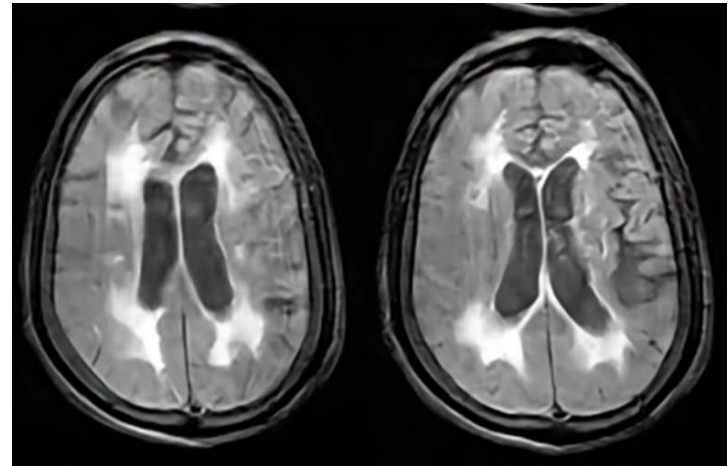
Vascular Parkinsonism

Scenario

- A patient with vascular risk factors with multiple small strokes and stepwise worsening

Clinical Features

- Symmetrical parkinsonism (rigidity/bradykinesia)
- Presents as “lower body parkinsonism” i.e. arms relatively normal, but legs rigid and slow
- Falls common
- Pyramidal signs
- Many have dementia and incontinence (looks like NPH)
- Tremor is not common
- Poor response to levodopa



Multi-System Atrophy (MSA)



Clinical Features

- Symmetrical parkinsonism (rigidity/bradykinesia)
- No tremor
- Early falls
- Dysautonomia (orthostasis, erectile dysfunction, incontinence)
- Ataxia (in MSA-C)
- Pyramidal signs
- Stridor
- Distal myoclonus
- No response to levodopa (produces orofacial dyskinesia)

Progressive Supranuclear Palsy (PSP)

Clinical Features

- Symmetrical parkinsonism (rigidity/bradykinesia)
- Axial rigidity more than limb
- No tremor
- Early falls
- Vertical supranuclear gaze paresis
- Hyperfrontalis (wide-eyed stare)
- Severe dysarthria
- Mild dementia (frontal lobe dysfxn)
- Poor response to levodopa



Cortico-Basal Degeneration (CBD)

Clinical Features

- Markedly asymmetric parkinsonism (rigidity/bradykinesia)
- Dystonia
- Myoclonus (action or tactile stimulated)
- Apraxia
- Aphasia
- Cortical sensory loss
- Alien limb phenomenon
- No response to levodopa

Dementia with Lewy Bodies (DLB)

Scenario

- Patient that appears parkinsonian and has a dementia, gets an antipsychotic for hallucinations and gets much worse

Clinical Features

- Symmetric parkinsonism (rigidity, bradykinesia)
- Can have tremor
- Early dementia (frontal and visuospatial)
- Fluctuating attention “good days and bad days”
- Visual hallucinations
- Hyper-sensitive to antipsychotics
- Some response to levodopa (but this can worsen visual hallucinations)
- REM Behaviour Sleep Disorder

See Geriatrics Lecture for more!

MCQ #3 – 2026

A 68yo patient, from a nursing home, with a history of depression presents to hospital with symmetrical parkinsonism including rigidity and bradykinesia 3 months after a laproscopic cholecystectomy. What is the most likely culprit?

- a) escitalopram
- b) venlafaxine
- c) clozapine
- d) metoclopramide
- e) domperidone

Answer: D – Metoclopramide was likely started after surgery and then never stopped. Domperidone is safe in PD (acts peripherally), escitalopram and venlafaxine (for depression) could cause PD but rare cases, clozapine likely not being used for garden variety depression.

Peripheral Neuropathy



Approach to Peripheral Neuropathy

A brief word about terminology...

- **Mononeuropathy** = a disease of a single nerve
e.g. carpal tunnel syndrome, ulnar neuropathy
- **Peripheral Neuropathy** = any disease of two or more nerves
e.g. polyneuropathy, polyradiculopathy, mononeuropathy multiplex
 - **Polyneuropathy** = a subset of peripheral neuropathy, syndrome of symmetrical, length-dependent signs/symptoms
 - e.g. diabetic polyneuropathy
 - **Polyradiculopathy** = a subset of peripheral neuropathy, disease of the nerve roots
 - e.g. GBS (AIDP)
 - **Mononeuropathy Multiplex** = a subset of peripheral neuropathy, disease of multiple, individual, named nerves
 - e.g. mononeuritis multiplex of vasculitis

Approach to Neuropathy

72M presents with a two year history of progressive bilateral foot numbness

1. Is it a peripheral neuropathy?

Consider other localizations:

- CN involved, unilateral findings, ataxia, cortical signs (aphasia, neglect) = brain
- bowel/bladder involvement, crossed findings = spinal cord
- pure motor findings = motor neuron disease, myopathy, NMJ

2. Temporal course

- ?acute, subacute, **chronic**

3. Pattern

- focal, multifocal, diffuse but non-length dependent, **diffuse: length dependent, symmetrical**

4. Fibre type

- **large fibre sensory, small fibre sensory**, autonomic, motor

5. Type of nerve pathology via EMG/NCS testing : **Axonal** or Demyelinating

6. Family History? Yes or **No**

“Garden variety Polyneuropathy”



(Eg) 72M presents with a two year history of progressive bilateral foot numbness

1. Is it a peripheral neuropathy? (History and exam per previous slide)
2. Temporal course
 - **Chronic** (months to years)
 - Tingling precedes numbness. Sensory symptoms predominate over motor initially.
 - Cramps precede motor weakness
3. Pattern
 - **diffuse: length dependent, symmetrical** [“stocking then glove distribution”]
4. Fibre type
 - **large fibre sensory>small fibre sensory**
 - **sensory > motor**
5. Type of nerve pathology via EMG/NCS testing : **Axonal**
6. Family History? **No**

Approach to Polyneuropathy



The simple DDx of the “**usual, common**” polyneuropathy

1. Idiopathic Sensory Neuropathy of the Elderly aka ISNE (>60yo)
2. Metabolic/Endocrine -
 - diabetes
 - kidney/liver disease
 - B12 deficiency
 - thyroid
3. Toxic
 - Alcohol
 - Chemotherapy (-platins, vinca alkaloids, taxanes)
 - Many others (see next slide)
4. Far less commonly, consider if clinical suspicion only based upon overall picture - vasculitis, infections (HIV, Hep C if cryoglobulinemia, Lyme)

Self Study – Approach to Polyneuropathy

Symmetrical distal sensorimotor

Acromegaly	Liver disease
Alcohol use	Malignancy
Amyloidosis	Medication induced (see Table 2)
Celiac disease	Nutritional deficiencies (vitamins B ₆ , B ₁₂ , and E; thiamine; folate; copper; phosphate)
Chronic kidney disease	Postgastrectomy syndrome
Connective tissue diseases	Toxin exposure (heavy metals, carbon monoxide, acrylamide, hexacarbons, ethylene oxide, glue)
Cryoglobulinemia	Vasculitis
Diabetes	Whipple disease
Gouty neuropathy	
Hypothyroidism/hyperthyroidism	
Infections (HIV/AIDS, Lyme disease)	
Inherited diseases (Charcot-Marie-Tooth disease, familial amyloidosis)	

Symptoms

Distal sensory and motor symptoms involve both sides symmetrically
 Sensory symptoms include pain, paresthesias, vibration, and proprioception
 Motor symptoms may include weakness and atrophy
 Autonomic symptoms may be present

TABLE 2

Medications and Toxins That May Cause Peripheral Neuropathy

Antiepileptic drugs	Chemotherapy agents	Metals
Lithium	Bortezomib (Velcade)	Arsenic
Phenytoin (Dilantin)	Cisplatin	Gold
Antimicrobial/antiviral drugs	Epothilones	Lead
Chloroquine	Oxaliplatin	Mercury
Dapsone	Paclitaxel	Thallium
Didanosine	Thalidomide	Solvents
Ethambutol	Vincristine	Acrylamide
Interferon alfa	Other drugs	Carbon disulfide
Isoniazid	Amitriptyline	Carbon monoxide
Metronidazole (Flagyl)	Cimetidine	2,4-dichlorophenoxy-acetic acid
Nitrofurantoin	Colchicine	Ethylene oxide
Stavudine	Disulfiram (Antabuse)	Glue
Triazoles	Levodopa	Hexacarbons
Cardiovascular drugs	Misoprostol (Cytotec)	Organophosphorus esters
Amiodarone	Nitrous oxide	
Hydralazine	Pyridoxine	
Procainamide	Suramin	
Statins	Tumor necrosis factor alpha inhibitors	

Approach to Polyneuropathy



Investigations of the “usual, common” polyneuropathy

- Bloodwork
 - CBC
 - Cr
 - HbA1c (if A1c normal, check OGTT)
 - B12
 - TSH
 - SPEP with immunofixation
- EMC/NCS - to characterize fibre-type, ensure axonal and not demyelinating

Approach to Polyneuropathy



Treatment of the “**usual, common**” polyneuropathy

- treat the cause (diabetes, B12 deficiency, renal failure etc)
- if neuropathic pain (burning, sharp, electrical pain), can try a variety of oral treatments (see next slide)
- there is no treatment for numbness

“Garden Variety Polyneuropathy”



Oral & Topical Treatment of Painful Diabetic Polyneuropathy (AAN Guideline Dec 2021)

- When initiating treatment, goal of therapy is reduction (~30%) of pain (elimination of pain will be challenging)
 - 4 classes of drugs have evidence
 - TCAs - amitriptyline
 - SNRIs - duloxetine
 - Gabapentinoids - pregabalin, gabapentin
 - Na Channel Blockers - carbamazepine, valproic acid
- **Opioids should NOT be used**
- Medication failure
 - Intolerable side effects OR
 - inadequate response after a demonstrated efficacious dose is tried for 12 weeks
- If patients prefer topical treatments (capsaicin), these can be tried though evidence is weaker

“Other Polyneuropathy”

If motor features predominate, acute/subacute history consider EMG more urgently.

If chronic, insidious and Family History - consider genetic neuropathies

If Demyelinating pattern Ddx

- **Inflammatory**

- Acute inflammatory demyelinating polyneuropathy (AIDP) aka Guillain Barre Syndrome
- subacute or chronic - Chronic inflammatory demyelinating polyneuropathy (CIDP)

- **Hereditary**

- Charcot Marie Tooth - 1/2500 people, far more common than paraneoplastic and inflammatory neuropathies. Variable penetrance and many variants (including demyelinating variant) beyond scope of GIM knowledge. Clues:
 - +ve family history
 - onset in childhood (‘tripping’, ‘not athletic’, ‘clumsy’)
 - insidious, symmetrical, slow to progress
 - severe weakness, yet surprisingly minimal disability (the patient learns to compensate well b/c the disease progresses slowly)
 - “negative” sensory symptoms (numbness) BUT **no** “positive” sensory symptoms (burning/tingling/electrical pain)
 - skeletal deformities = hammertoe, pes cavus

CIDP is a good “GIM scenario” -

DDx includes many systemic disorders:

Infection - Lyme Hep B/C, HIV

Autoimmune - SLE, IBD, thyroid

Paraneoplastic (paraproteinemias especially)

For workup + mgmt of AIDP see Neuro Emergencies lecture



ALS

(If time allows! If not – self-study)



Amyotrophic lateral sclerosis (ALS)

Neurodegenerative disease leading to selective loss of pyramidal neurons (**primary motor cortex**) & motor neurons/anterior horn cells (**spinal cord**)

- Patients typically die from respiratory failure within 5 years of diagnosis
- ALS eventually involves multiple levels of the neuraxis:
 - Bulbar (pseudobulbar affect, dysphagia, dysarthria, dyspnea, brisk jaw jerk)
 - Cervical, thoracic, and lumbar levels
 - Expect lower motor findings at the level of the lesion (atrophy, fasciculations, weakness) and upper motor neuron findings below the lesion (spasticity, hyper-reflexia, upgoing toes)

ALS: clinical clues

Clinical differentiation between cervical myelopathy (common, due to spondylosis)

History:

- Progressive symptoms: onset of asymmetric limb weakness in 75%, bulbar symptoms in 25%
- Cranial nerve dysfunction (dysphagia, dysarthria) above C-spine
- Fronto-temporal degeneration - cog / behavioural decline in up to 50%
- Family history of motor neuron disease (10% of cases)

Physical:

- Pseudobulbar affect
- Cranial nerve findings (tongue wasting, fasciculations, weakness, brisk jaw jerk)

ALS: 2020 Cdn Best Practice Recommendations

- Work-up
 - MR cervical spine to exclude structural lesion
 - NCS/EMG
- CMAJ 2020: **ALS diagnosis should be confirmed by an expert clinic within 4 weeks.**
- Treatment - **disease modifying therapies** in Canada
 - Multidisciplinary Clinics have great benefit: ↑survival, ↓hospitalization, ↑QOL
 - **Riluzole** - ↑**survival** median 3 months, initiate after diagnosis of ALS
 - side effect: fatigue - consider reduce dose or discontinue if disabling
 - **Edaravone** - ↓**functional decline** in select subgroup of patients (beyond scope of GIM)
- Treatment - symptom relief:
 - Sialorrhea - anticholinergics, oral suctioning or botox
 - Spasticity - physio/stretching, baclofen, tizanidine, botox, BZD, CBD could be considered
 - Cramps - tonic water, gabapentin, baclofen
 - Depression / anxiety - SSRI or SNRI first line along with non pharm support. BZD will exacerbate breathing difficulties

ALS: 2020 Canadian BPG: Nonpharmacologic Mgmt

- Respiratory care - FVC @ baseline, q3 months + 1+ of SNIP, supine FVC or MIP
 - Home NIV [criteria below], Airway clearance management
 - Symptoms - orthopnea
 - SNIP <40cmH₂O or MIP <40cm H₂O
 - Upright FVC <65%
 - FVC sitting or supine <80% with symptoms or signs of resp insufficiency
 - Daytime hypercapnia pCO₂ >45mmHg
 - AbN nocturnal oximetry or symptomatic sleep disordered breathing
- VTE risk - higher in pts w leg onset ALS. Thromboprophylaxis has not been studied.
- Nutritional Considerations - BMI @ Baseline, q 3months
 - High calorie, high carb diets
 - Consider early enteral tube insertion if:
 - aspiration risk / bulbar symptoms / dysphagia
 - 5-10% reduction weight, >= 1 pt reduction BMI or BMI <18.5
 - ↓FVC approaching 50% even in absence of dysphagia
- QOL and end of life Considerations: early Palliative Care referral, caregiver support, discuss MAID, organ donation

MCQ option in ALS:

“Pseudobulbar palsy”

- BULB = medulla
- PYRAMID = corticospinal tract
- Pseudobulbar palsy = **UMN lesion in corticobulbar pathway**
 - Corticobulbar path → synapse @ nuclei of CN V, VII, IX, X, XI, XII
 - PRESENTATION:
 - Dysphagia
 - Dysarthria
 - UMN findings (hyperactive jaw jerk)
 - Pseudobulbar affect
- Can be caused by a variety of CNS lesions: ALS, Multiple systems atrophy, Progressive supranuclear palsy...

Thank You!



Bonus Slides

- Neuro-ophthalmology
 - **check out 2022 video online**
- Peripheral Neuropathy study slides
 - **check out 2018 video online**
- Stroke Trials
 - For your reference, good to know, but not NEED to know for exam
- Lumbar Puncture Counselling
- COVID & Neurology

Neuroanatomy Review

Online Video from IMR2024 available for those who need a refresher!

Neuro-Ophthalmology

These slides are presented by Dr Atia and available online on both CHYME and internalmedicinereview.ca

“Neuro-ophthalmology for the internist”

HIGHLY RECOMMENDED VIEWING

Please see the 2018 Video on our website “DIPLOPIA” for an ophthalmologist’s approach to this tricky subject!

MCQ

29F in ER with T1DM presents with right ptosis and diplopia. When examined, visual acuity is 20/20 in each eye, the left eye is slightly adducted. She reports diplopia when looking to the left but also when looking down and in. The pupils are equal and responsive to light. CT Head is normal. Four months prior, she had right ptosis and diplopia and found to have a HbA1c of 7.4%.

What is the likeliest localization and etiology of the current symptoms?

- a) right 3rd nerve and left 6th nerve, ischemic/diabetic nerve
- b) left 3rd nerve and right 6th nerve, ischemic/diabetic nerve
- c) right 3rd nerve and left 6th nerve, carotid cavernous fistula
- d) left 3rd nerve and right 6th nerve, carotid cavernous fistula
- e) right 3rd nerve and left 6th nerve, myasthenia gravis
- f) neuromuscular junction, myasthenia gravis

Approach to Ophthalmoscopy

How to use an ophthalmoscope like you know what you're doing even though you may not...

1. turn off the lights in the room
2. tell the pt what you will be doing
3. use the largest diameter light on the ophthalmoscope
4. ask the pt to stare at a target directly ahead
5. use your **right** hand and your **right** eye to look at pts **right** eye, then use your **left** hand and your **left** eye to look at pts **left** eye
6. start at a distance and comment on red reflex being present
7. then move in slowly and aim slightly (10 degrees) "up and in" and adjust the focus to focus on the optic disc
8. comment on the optic disc (sharp margins, not blurred) and optic nerve colour (no pallour) and vessels (no clot seen, spontaneous venous pulsations)
9. look around retina and comment on other abnormalities
10. ask pt to look directly into the light, allowing you to then see the fovea/macula (comment that there is no cherry red spot)

Approach to Ophthalmoscopy

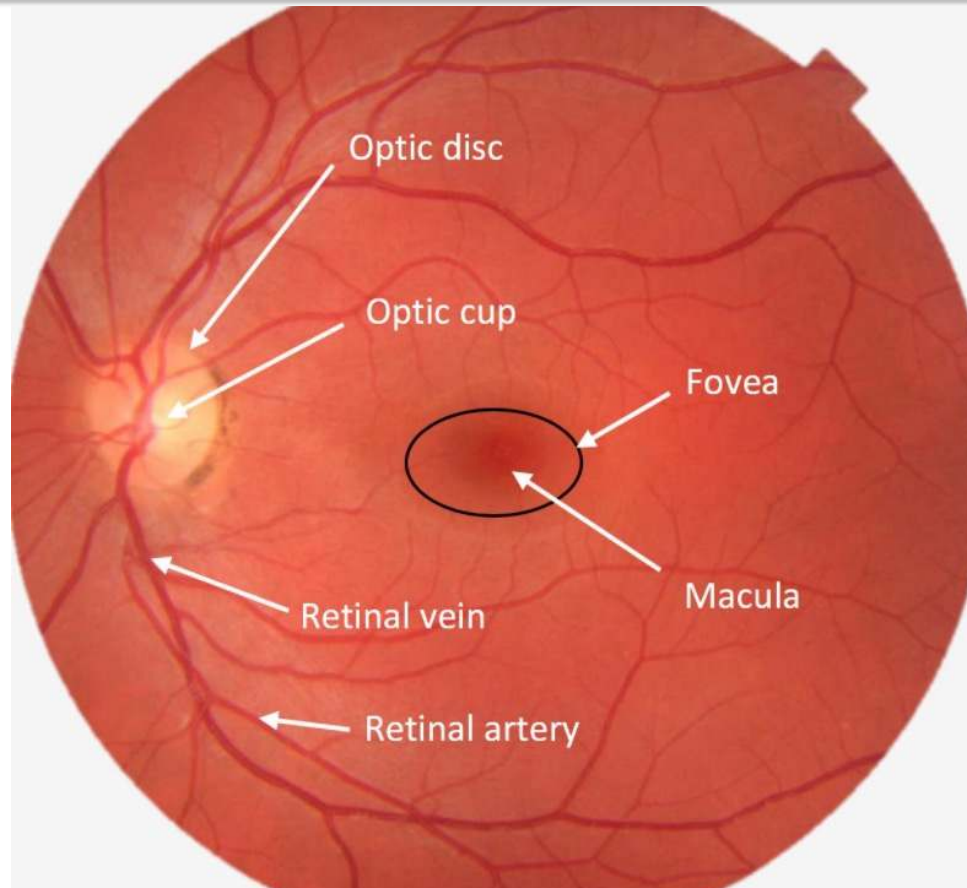
Poor form:



Good form:



Approach to Ophthalmoscopy

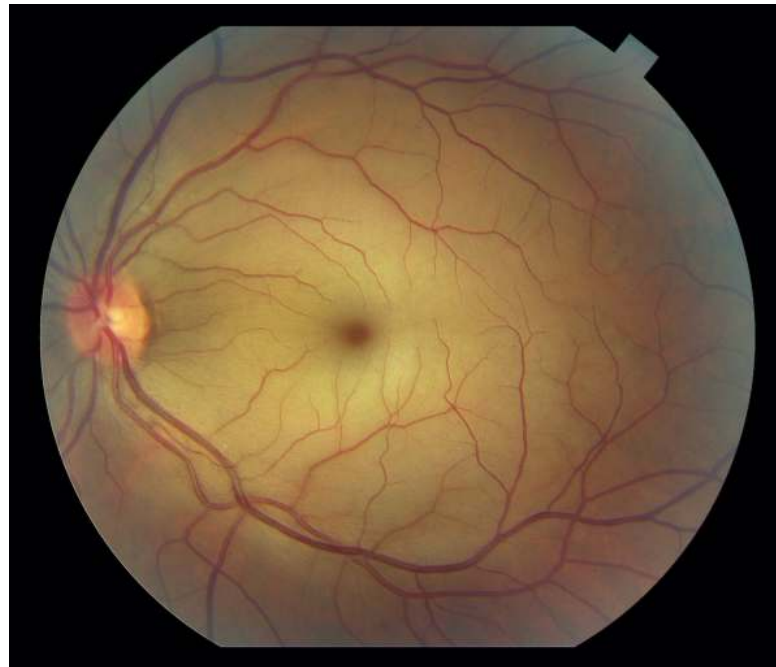
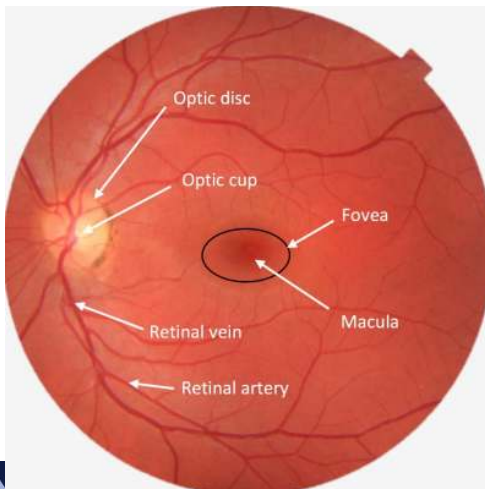


Approach to Ophthalmoscopy

Scenario: vasculopath with sudden, complete, painless unilateral vision loss

Central Retinal Artery Occlusion with red spot cherry

Normal

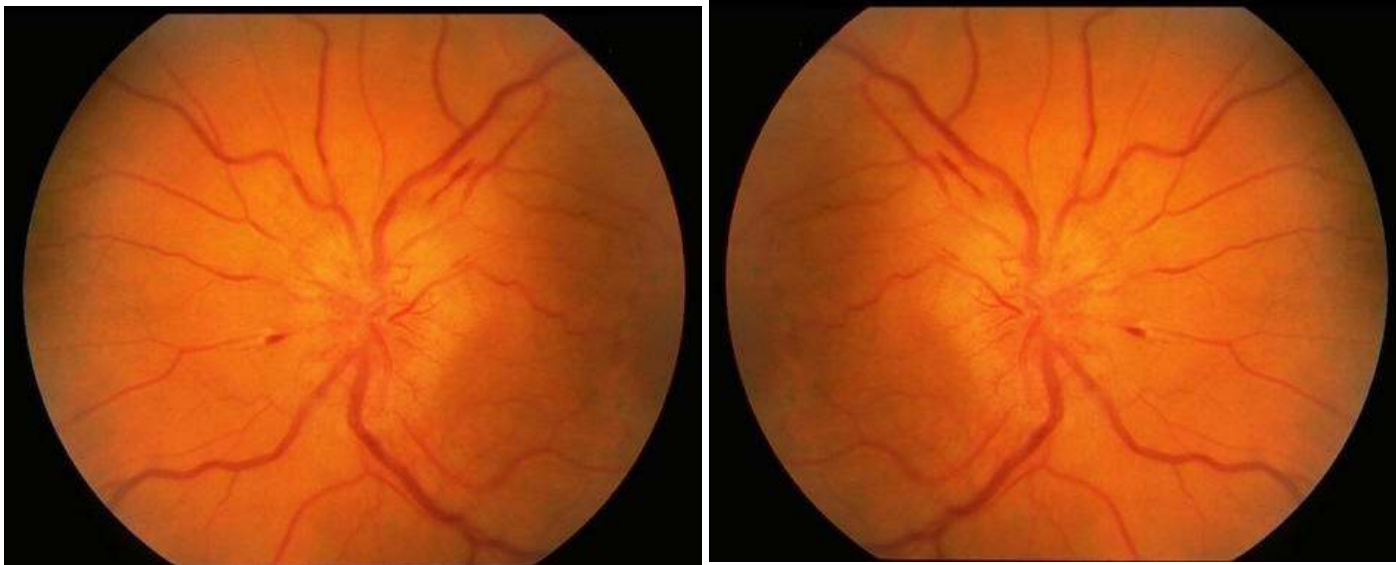
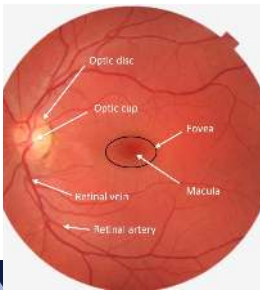


Approach to Ophthalmoscopy

Scenario: pt with lung cancer and new severe morning headaches

Bilateral disc edema = likely papilledema

Normal

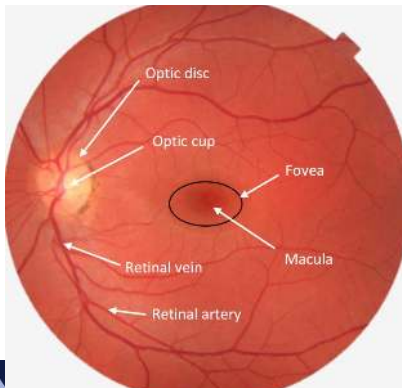


Approach to Ophthalmoscopy

Scenario: young female, unilateral eye pain and vision loss

disc edema = Acute Optic Neuritis

Normal

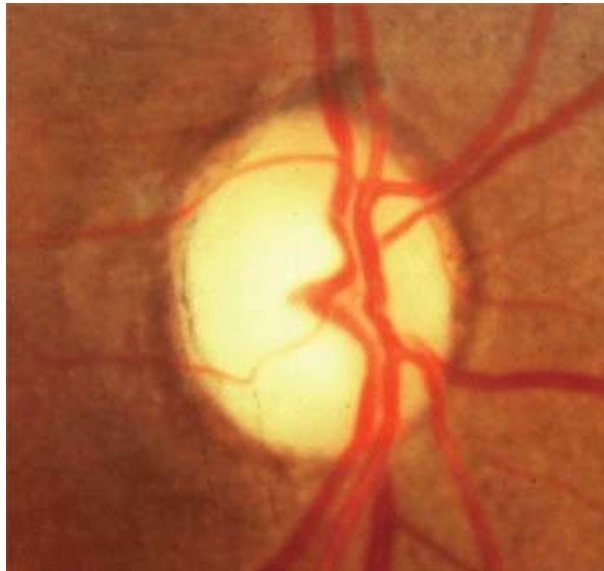
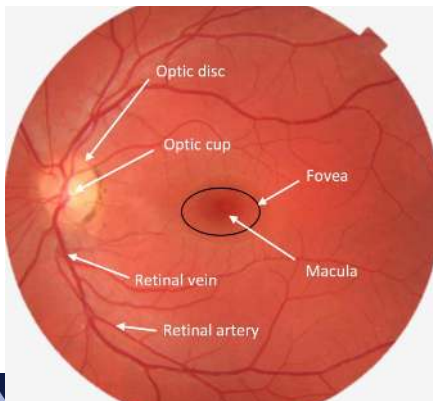


Approach to Ophthalmoscopy

Scenario: same young female, vision loss has mostly recovered, now examining few months later

pale disc = Prior Optic Neuritis

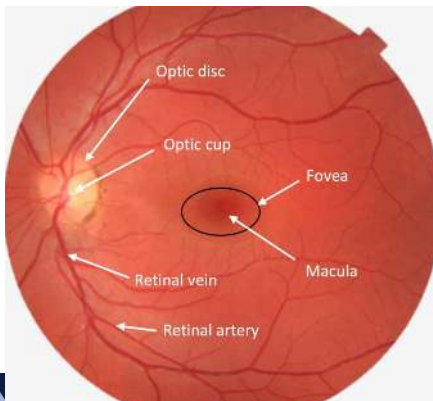
Normal



Approach to Ophthalmoscopy

Scenario: older person (>60), poor peripheral vision, no pain, eye pressure elevated
large cup-to-disc ratio and pale disc = glaucoma

Normal

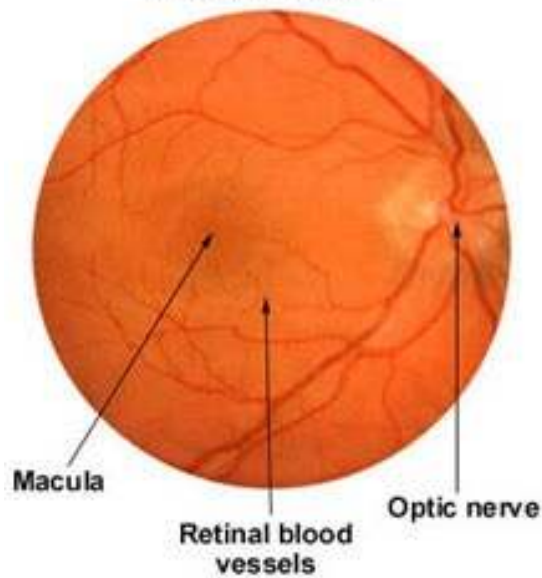


Approach to Ophthalmoscopy

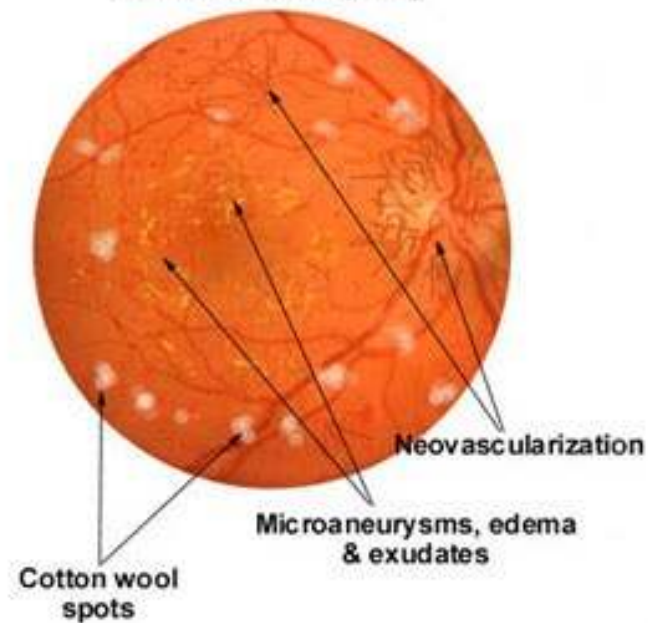
Scenario: diabetic patient

cotton wool spots, **microaneurysms**, hard exudates, **neovascularization**, flame hemorrhages = diabetic retinopathy

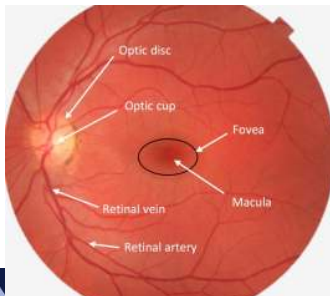
Normal Retina



Diabetic Retinopathy



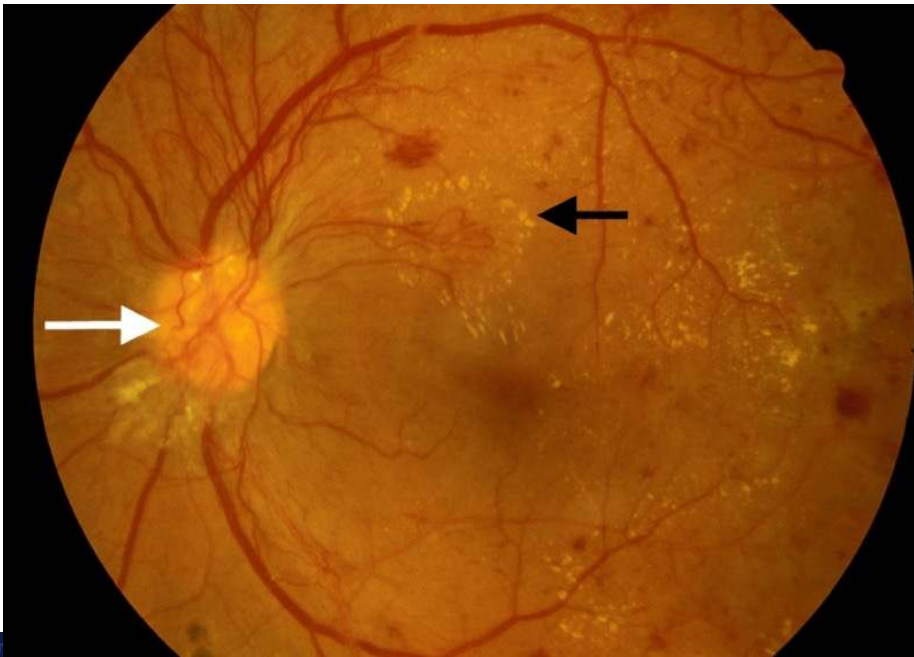
Normal



Approach to Ophthalmoscopy

Scenario: diabetic patient

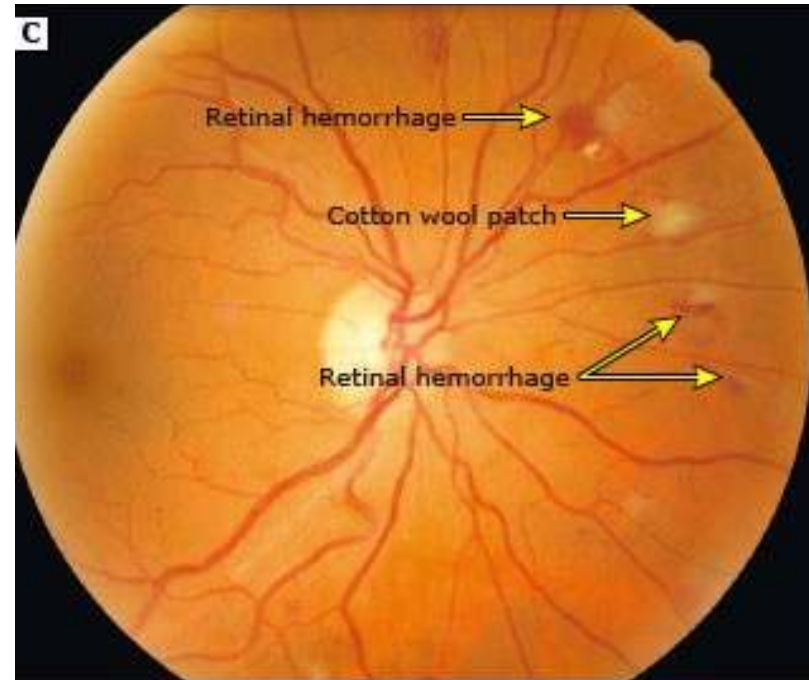
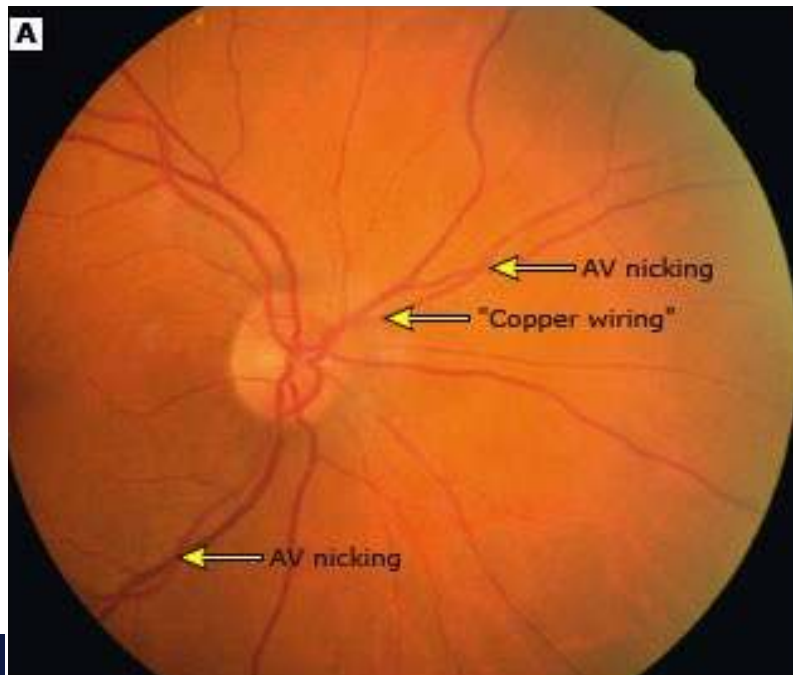
cotton wool spots, **microaneurysms**, hard exudates, **neovascularization**, flame hemorrhages = diabetic retinopathy



Approach to Ophthalmoscopy

Scenario: hypertensive patient

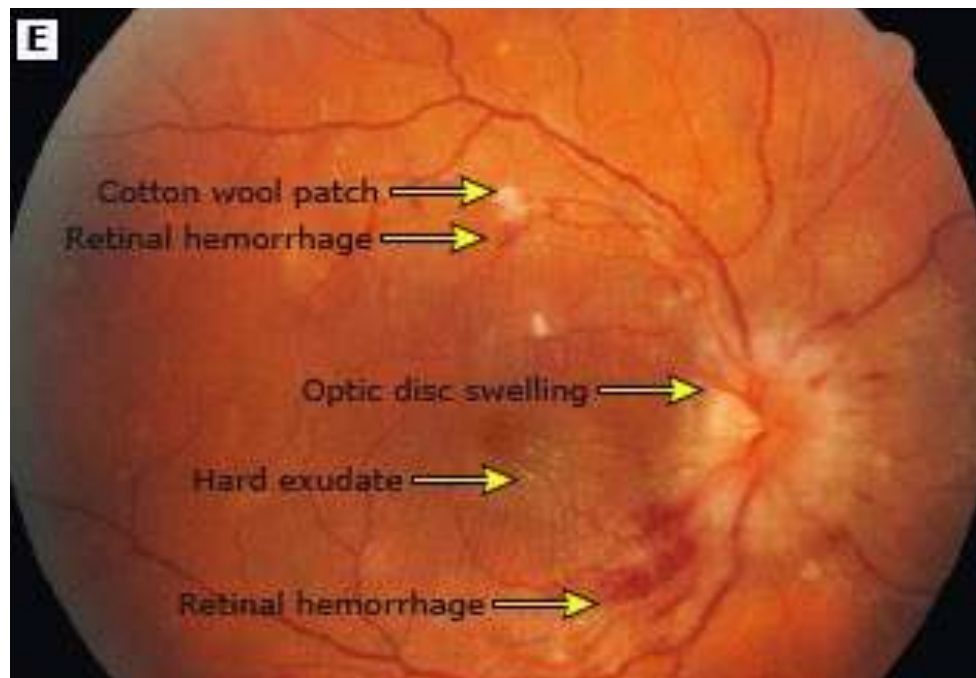
cotton wool spots, hard exudates, flame hemorrhages, **AV nicking**, **copper wiring**,
disc edema = hypertensive retinopathy



Approach to Ophthalmoscopy

Scenario: hypertensive patient

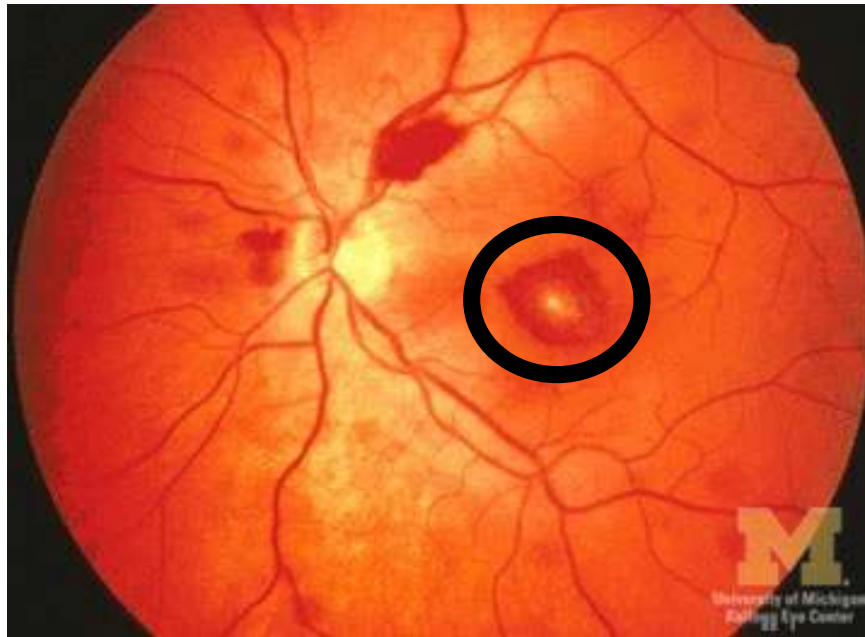
cotton wool spots, hard exudates, flame hemorrhages, AV nicking, copper wiring,
disc edema = hypertensive retinopathy



Approach to Ophthalmoscopy

Scenario: fever, new cardiac murmur

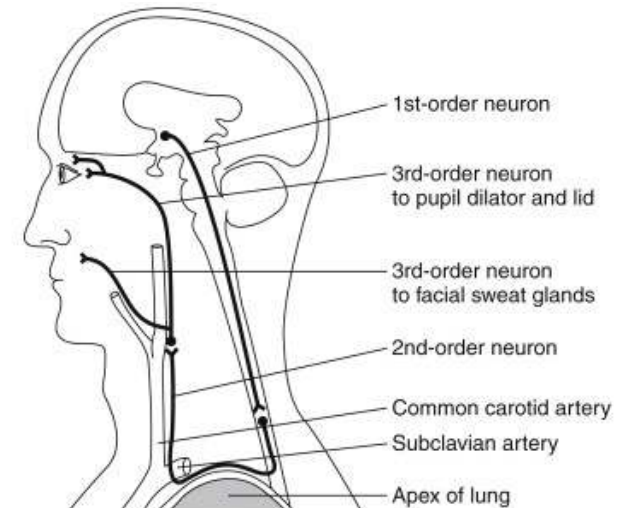
roth spot = infective endocarditis



Approach to Diplopia

Relevant Anatomy

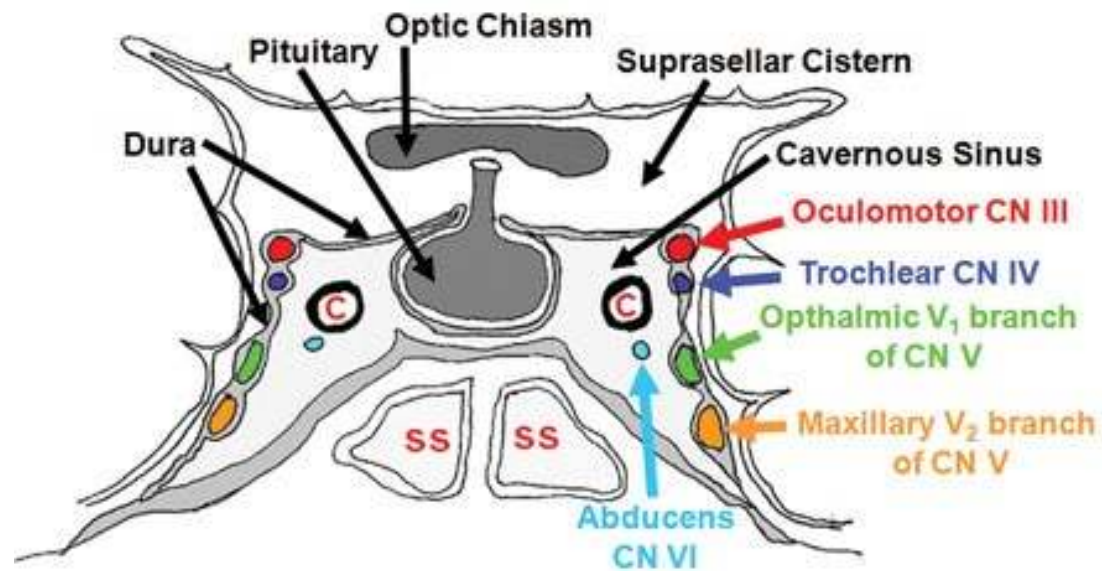
1. 6th n. - LR
2. 4th n. - SO
3. 3rd n. - MR, IR, SR, IO
 - + Levator Palpebrae Superioris (elevates eye lid)
 - + Parasympathetics (cause pupil constriction)
4. Sympathetics
 - (cause pupil dilation & controls Muller's muscle which also elevates eyelid)



Approach to Diplopia

Relevant Anatomy

Cavernous Sinus : CN III, IV, V₁, V₂, VI and sympathetics



Approach to Diplopia

The 5 Question History

1. Monocular or Binocular diplopia?
 1. monocular = psych or ophtho
 2. binocular = neurology
2. Images horizontal, vertical, or oblique to each other?
 1. horizontal = 6th n. palsy*
 2. vertical = 4th n. palsy*
 3. oblique = 3rd n. palsy*
3. Diplopia worse looking near or far?
 1. near = 3rd or 4th n. palsy
 2. far = 6th n. palsy
4. Diplopia worse with looking up or down or left or right?
 1. left or right = 6th n. palsy
 2. up or down = 3rd or 4th n. palsy
5. Associated Symptoms (ptosis, eye pain, headache, vision loss, recent trauma, other cranial nerve involvement, fluctuations)

Approach to Diplopia

- For all the following 7 cases, the patient is asked to look straight ahead (primary position)
- ANSWER KEY IS BELOW... DON'T PEEK ;)

				Answer Key: DON'T CHEAT!
				Horner's on L
				Horner's on R
				3rd N palsy on L, pupil involved
				3rd N palsy on R, pupil spared
				6th N palsy on R
				4th N palsy on L
				R 7th N (bell's) palsy

Case 1

Localization?



Case 2

Localization?



Case 3

Localization?



Case 4

Localization?



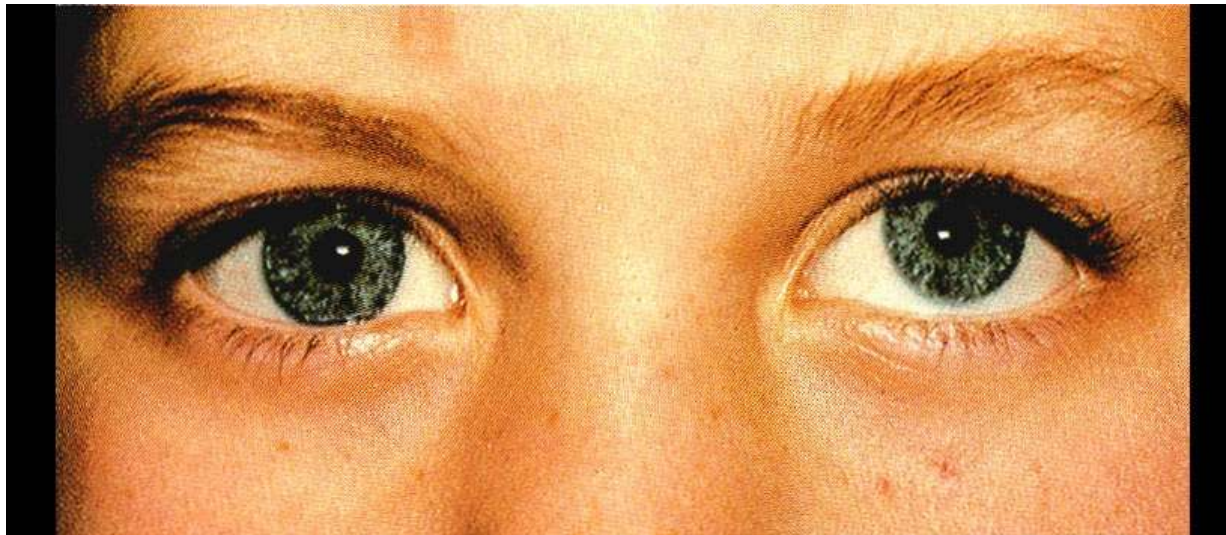
Case 5

Localization?



Case 6

Localization?



Case 7

Localization?



Approach to Diplopia

	3rd n. palsy	Horner's
ptosis	++	+
pupil	possible mydriasis (dilated)	miosis (constricted)
anisocoria	worse in light	worse in dark
diplopia (EOM abnormality)	yes (down and out)	no
must r/o	PCOMM aneurysm	carotid dissection (in neck)

Approach to Diplopia

Possible Localizations (and common etiologies) in Binocular Diplopia

1. Extraocular Muscle (thyroid eye disease)
2. NMJ (MG)
3. Nerve, 3rd, 4th, 6th anywhere incl. cavernous sinus
(ischemic, compressive, infectious, inflammatory, infiltrative etc)
4. Nucleus, 3rd, 4th, 6th in brainstem (stroke, MS)
5. Internuclear Pathways in brainstem
(stroke, MS causing an INO)

3rd nerve palsy

	ischemic 3rd n. palsy	compressive 3rd n. palsy
EOM	down and out	
ptosis	+	
pupil	spared	involved (mydriasis, poor response to light)
pain	minimal/no	yes
context	DM, HTN	Aneurysm

When should you do a CTA?

- 1) complete 3rd n. palsy and pupil spared = no need for CTA
- 2) partial 3rd n. palsy and pupil spared = must do CTA
- 3) pupil involved = must do CTA

Horner's Syndrome

Triad:

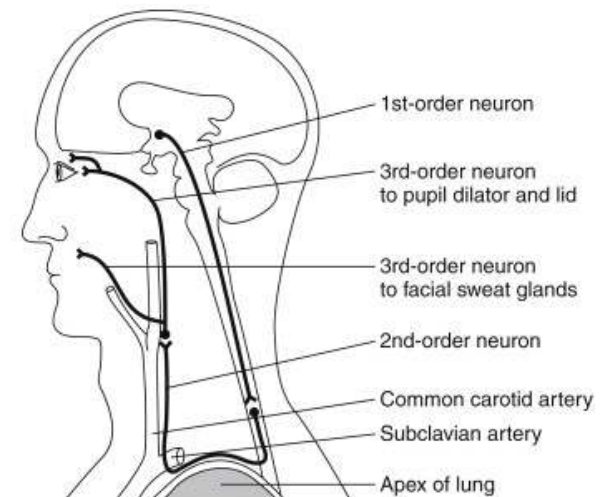
1. Ptosis
2. Miosis
3. Anhidrosis (if first or second order neuron is damaged)

DDx depends on localization:

1. 1st order neuron - stroke, demyel, tumour
2. 2nd order neuron - T1 radic, pancoast tumour
3. 3rd order neuron - carotid dissection/aneurysm

Investigations:

1. 1st order neuron - MRI Brain
2. 2nd order neuron - MRI C-spine/CT Chest
3. 3rd order neuron - CTA neck and brain



6th nerve palsy

Clinical Presentation

- binocular diplopia, horizontal images, worse when looking far, worse when looking left (if left CN6 palsy) or right (if right CN6 palsy)
- Possible Localization (and etiologies)
 - brainstem (stroke, demyelination, tumour)
 - cavernous sinus (fistula, aneurysm, dissection, pituitary tumour, inflammation, infection)
 - false-localizing (inc ICP from tumour, VST, **IIH**, **trauma**)
 - anywhere along course of nerve (**ischemia**)
- Investigations
 - CT/CTA intracranial vessels (for cavernous sinus pathology)
 - MRI Brain with GAD (for stroke/tumour/inflammation/infection)
 - MRI Brain with MRV or CTV (for VST), then LP (for IIH)

MCQ - Answer

29F in ER with T1DM presents with right ptosis and diplopia. When examined, visual acuity is 20/20 in each eye, the left eye is slightly adducted. She reports diplopia when looking to the left but also when looking down and in. The pupils are equal and responsive to light. CT Head is normal. Four months prior, she had right ptosis and diplopia and found to have a HbA1c of 7.4%.

What is the likeliest localization and etiology of the current symptoms?

- a) right 3rd nerve and left 6th nerve, ischemic/diabetic nerve
- b) left 3rd nerve and right 6th nerve, ischemic/diabetic nerve
- c) right 3rd nerve and left 6th nerve, carotid cavernous fistula
- d) left 3rd nerve and right 6th nerve, carotid cavernous fistula
- e) right 3rd nerve and left 6th nerve, myasthenia gravis
- f) **neuromuscular junction, myasthenia gravis**

answer f)

a) is a possible localization based on description but simultaneous right 3rd and left 6th nerve disease is less likely than NMJ disease (MG)
e) is tempting but MG is by definition an NMJ disease so f) is correct

Neuropathy - the physical exam

These slides for your REFERENCE.

We have comprehensive videos for the following on our website, in 2018 Videos Section:

Note that as of 2021 with Applied exams online the physical exam seems less of a focus for study purposes.

Physical Exam for POLYNEUROPATHY

Physical Exam for Nerve ROOT versus Peripheral Nerve lesions

C7 vs Radial

L5 vs Peroneal

C8/T1 vs Ulnar

MCQ

46F presents with a right foot drop. She has been renovating her bathroom recently and reports back pain. On exam, tone is normal throughout and strength is normal in the arms and left leg. Strength of right leg: hip flexion and extension 5, hip abduction and adduction 5, knee flexion and extension 5, ankle dorsiflexion 4, ankle plantar flexion 5, ankle eversion 4, ankle inversion 5. Reflexes are 2+ in all four limbs. Sensation to pin is reduced in the right lateral calf and dorsum of the foot and involving the lateral ankle.

What is the most likely localization?

- a) motor neuron disease (ALS)
- b) right L5 radiculopathy
- c) right sciatic neuropathy
- d) right common peroneal neuropathy at the knee
- e) thoracic spinal cord



Relevant Anatomy

- The Screening Motor Exam & it's anatomy

Arm Muscle	Nerve	Root
Deltoid	Axillary	C5
Biceps	Musculocutaneous	C5,6
Triceps	Radial (prox to spiral groove)	C7
Finger Extensors	Radial	C7
FDI	Ulnar	C8,T1
FPL	Median	C8, T1

- Covers every major nerve, every nerve root, and allows pyramidal pattern to be detected
- If weakness seen in any muscle, test further muscles depending on root vs nerve (more on this later)

Relevant Anatomy

- The Screening Motor Exam & it's anatomy

Leg Muscle	Nerve	Root
Hip Flex (iliopsoas)	Branch L2/3, Femoral	L2,3
Hip Ext (glut max)	Inferior Gluteal	S1
Knee Flex (hams)	Sciatic	S1
Knee Ext (quads)	Femoral	L2,3,4
ADF (tibialis anterior)	Peroneal	L5
APF (gastrocs)	Tibial	S1

- Covers every major nerve, every nerve root, and allows pyramidal pattern to be detected
- If weakness seen in any muscle, test further muscles depending on root vs nerve (more on this later)

Relevant Anatomy

- The Screening Motor Exam & it's anatomy

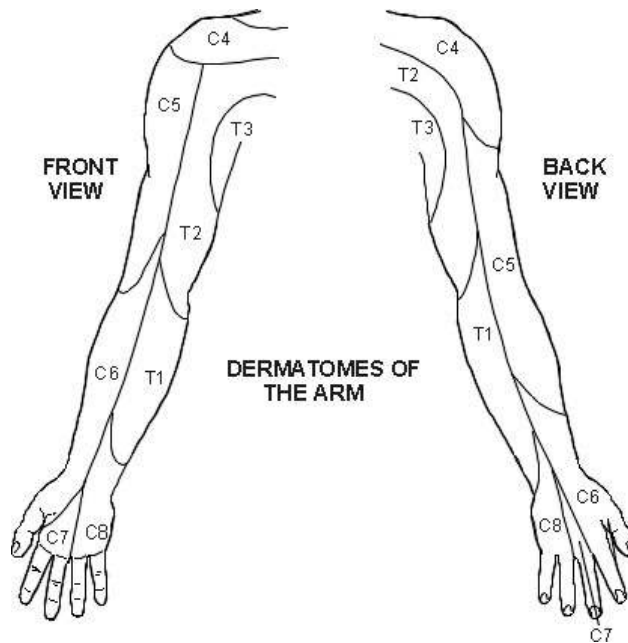
Reflexes	Nerve	Root
Biceps	Musculocutaneous	C5,6
Triceps	Radial (prox to spiral groove)	C7
Brachioradialis	Radial	C5,6
Knee/Patellar (quads)	Femoral	L2,3,4
Ankle/Achilles	Tibial	S1
Plantars	assesses UMN	

- Note: no reliable C8 or L5 reflex

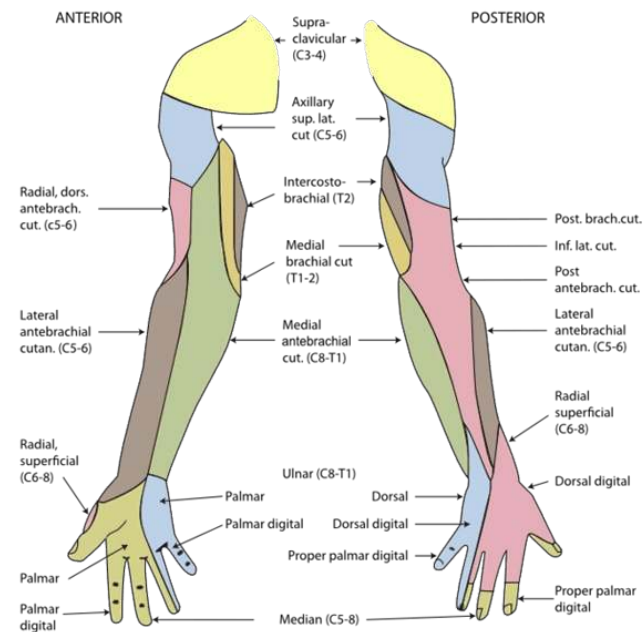
Relevant Anatomy

- The Screening Sensory Exam & it's anatomy

Roots/Dermatomes



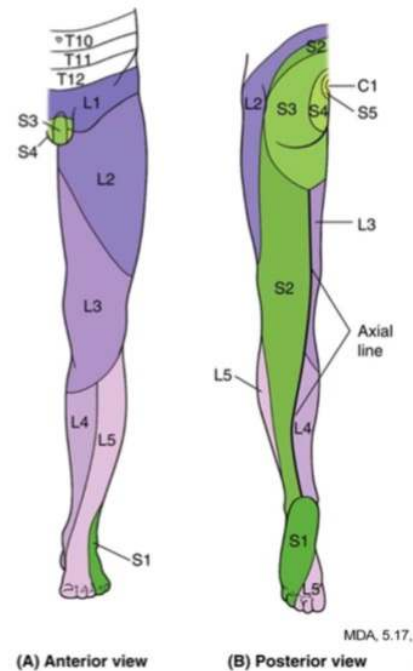
Peripheral Sensory Nerve



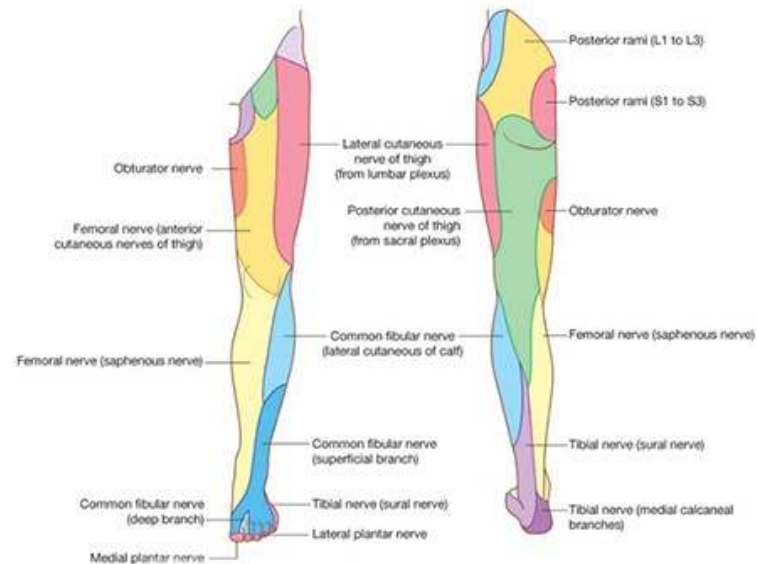
Relevant Anatomy

- The Screening Sensory Exam & it's anatomy

Roots/Dermatomes



Peripheral Sensory Nerve



Approach to Root vs Nerve

- Always use the following Grid:

	Exam	Nerve	Root
Motor - Power			
Motor - Reflex			
Sensory - Pin			
Special Tests			

- Practically, it is the **motor-power** exam that is most useful
- Memorize “back-pocket muscles” = key ones to assess root vs nerve

L5 vs Peroneal Neuropathy

Examination		Nerve	Root	in Peroneal <u>&</u> L5 lesion
Motor-Power	ADF Ankle Eversion	Peroneal	L5	weak
Sensory-Pin	Lateral Calf Dorsum Foot	Peroneal	L5	reduced

Examination		Nerve	Root	in Peroneal lesion	in L5 lesion
Motor-Power	Ankle Inversion	Tibial	L5	strong	weak
	Hip Abduction	Sup Gluteal	L5	strong	weak
Motor-Reflex	(Hamstrings)	Sciatic	L5	normal	absent
Sensory-Pin	Lateral Ankle	Peroneal	S1	reduced	normal
Special Tests	Tinel @ Fib Head	Peroneal	N/A	positive	negative

- Back-pocket muscle = **red**

MCQ - Answer

46F presents with a right foot drop. She has been renovating her bathroom recently and reports back pain. On exam, tone is normal throughout and strength is normal in the arms and left leg. Strength of right leg: hip flexion and extension 5, hip abduction and adduction 5, knee flexion and extension 5, ankle dorsiflexion 4, ankle plantar flexion 5, ankle eversion 4, ankle inversion 5. Reflexes are 2+ in all four limbs. Sensation to pin is reduced in the right lateral calf and dorsum of the foot and involving the lateral ankle.

What is the most likely localization?

- a) motor neuron disease (ALS)
- b) right L5 radiculopathy
- c) right sciatic neuropathy
- d) right common peroneal neuropathy at the knee**
- e) thoracic spinal cord



C7 vs Radial* Neuropathy

*assuming lesion @ spiral groove

Examination		Nerve	Root	in Radial <u>&</u> C7 lesion
Motor-Power	wrist extension finger extension	radial	C7	weak

Examination		Nerve	Root	in Radial lesion	in C7 lesion
Motor-Power	brachioradialis	Radial	C5,6	weak	strong
	pronator teres	Median	C7	strong	weak
	triceps	Radial ^{above spiral groove}	C7	strong	weak
Motor-Reflex	triceps	Radial ^{above spiral groove}	C7	normal	absent
	brachioradialis	Radial	C5,6	absent	normal
Sensory-Pin	middle finger	Median	C7	normal	absent
	dorsum hand	Radial	C6	absent	normal
Special Tests	Spurling's test	N/A	C7	negative	positive

- Back-pocket muscle = **red**

C8/T1 vs Ulnar Neuropathy

Examination		Nerve	Root	in Ulnar <u>&</u> C8/T1 lesion
Motor-Power	FDI	ulnar	C8/T1	weak
Sensory-Pin	4th & 5th digits	ulnar	C8/T1	absent

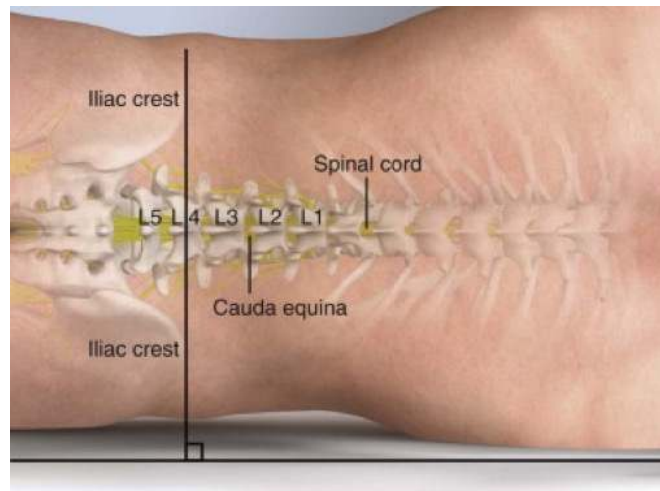
Examination		Nerve	Root	in Ulnar lesion	in C8/T1 lesion
Motor-Power	FPL	median (AIN)	C8/T1	strong	weak
	APB	median	C8/T1	strong	weak
	EIP	radial	C8	strong	weak
Motor-Reflex	none useful				
Sensory-Pin	none useful				
Special Tests	Spurling's test	N/A	C8/T1	negative	positive

- Back-pocket muscle = **red**

Lumbar Puncture: Also a 2018 Online Video!

Anatomy of the LP

- spinal cord ends at L1-L2
- Tuffier's line (imaginary line b/w iliac crests) at L4
- lateral decubitus - can measure opening pressure
- sitting - canNOT measure opening pressure



Lumbar Puncture

Anatomy of the CSF

- produced by the choroid plexus in the ventricles of brain
- total volume = 150cc
- production in one day = 500cc
- production in one hour = 20cc
- recall: you remove only 8-20cc in one LP

Lumbar Puncture

Contraindications

- INR > 1.7
- Plt < 50
- papilledema/raised ICP secondary to intracranial mass
- patient can't safely tolerate procedure

Pre-imaging Contraindications

- focal neuro deficit
- new seizure
- GCS<10 or Alt LOC that impairs neuro exam
- immunocompromised pt
- anatomical issues - spine instrumentation, tethered cord, chiari malformation, local skin infection

Lumbar Puncture

Consent:

Benefits/Purpose of the procedure

- potentially diagnostic

Risks of the procedure

- local discomfort
- infection $<1\%$
- hemorrhage $<1\%$
- nerve injury $<0.01\%$
- post-LP headache
 - 10-20% risk but vast majority recover within 48h
 - 4% of those with post-LP h/a will need blood patch

Lumbar Puncture

My Shotgun LP:

Opening Pressure

Tube 1: cell count and differential

Tube 2: glucose and protein, ACE level

Tube 3:

- BACTERIAL: gram stain, culture
- VIRAL: PCR (HSV, VZV, EBV, CMV, WNV, HHV-6, JC Virus, adenovirus, enterovirus) IgM for WNV
- MYCOBACTERIAL/OTHER: AFB, mycobacterial culture, VDRL, Lyme
- FUNGAL: culture, cryptococcus antigen

Tube 4: cytology and flow cytometry

Tube 5: cell count and differential

Imaging in Neurology



Approach to CT Head

1. patient's name + age
2. bone (bone window) - fractures
3. blood
 epidural/subdural/subarachnoid/intraparenchymal/
 intraventricular
4. ventricles
5. vessels
6. parenchyma - one hemisphere at a time

Always compare to prior imaging “old gold”

Approach to CT Head

Hyperdense/White on CT
1) Calcium/Bone
2) Contrast
3) Blood (acute, hours to days)
4) Melanoma & Lymphoma (because they often bleed, see #3)

Approach to CT Head

Blood on CT
min to hrs to days = hyperdense
1-2 weeks = isodense
3-4 weeks = hypodense

Plain CT rules out SAH if:

- 1) done within 12h from onset of bleed AND
- 2) read by neuroradiologist

In any other scenario, an LP must be done to rule out SAH.

Approach to MRI Brain

1. patient's name + age
2. T1
3. T2/FLAIR
4. DWI + ADC
5. GRE/SWI
6. T1 w GAD

Approach to MRI Brain

T1 sagittal

normal anatomy
pituitary gland
cerebellum
brainstem

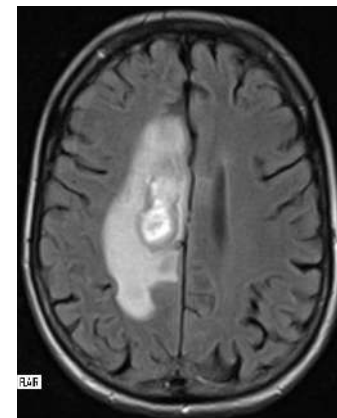
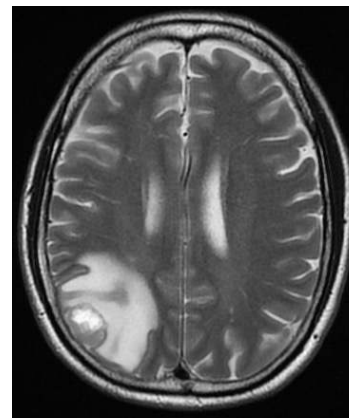
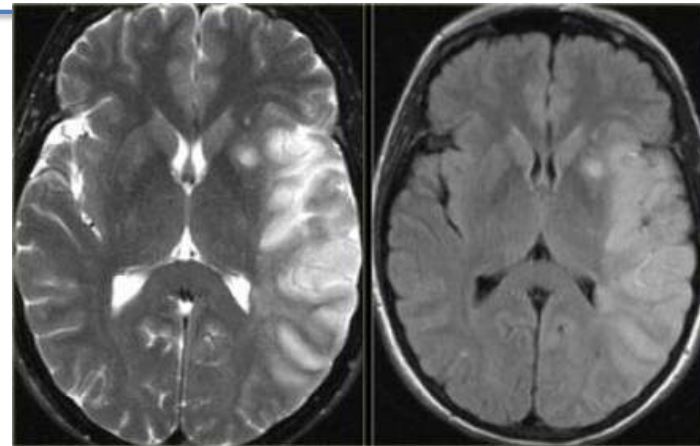


Approach to MRI Brain

T2 and FLAIR

any pathology = hyperintense

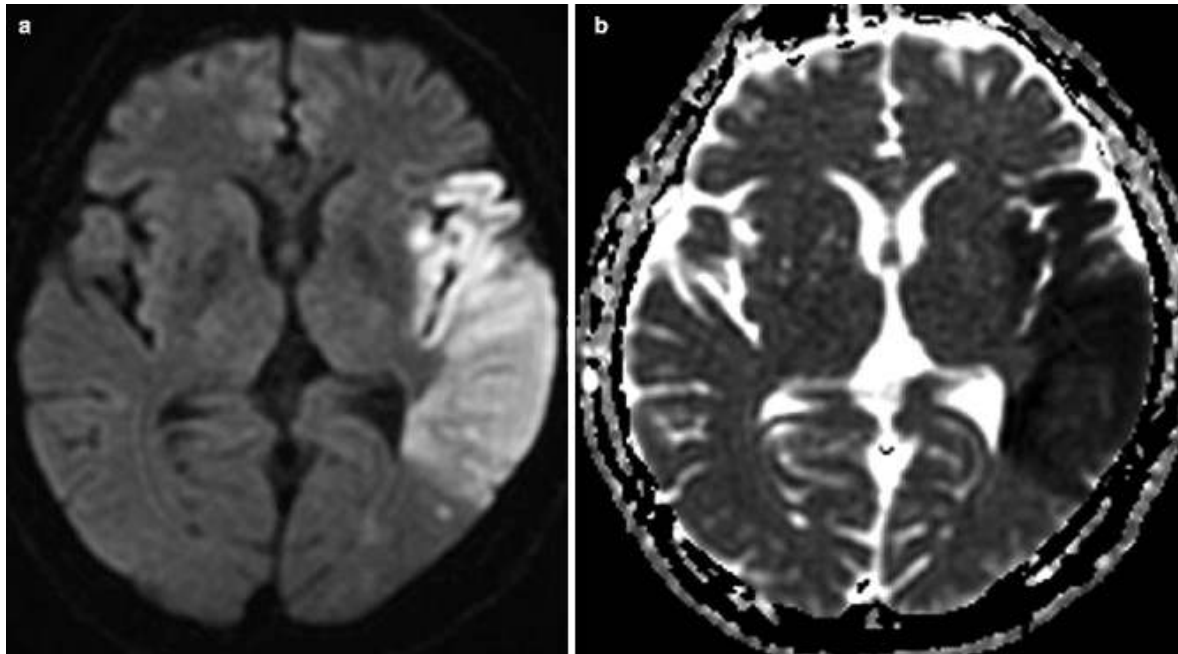
- stroke
- tumour
- blood
- demyelination



Approach to MRI Brain

DWI & ADC

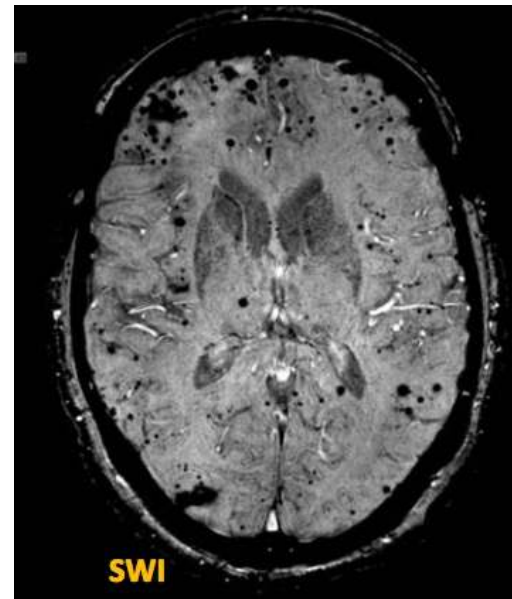
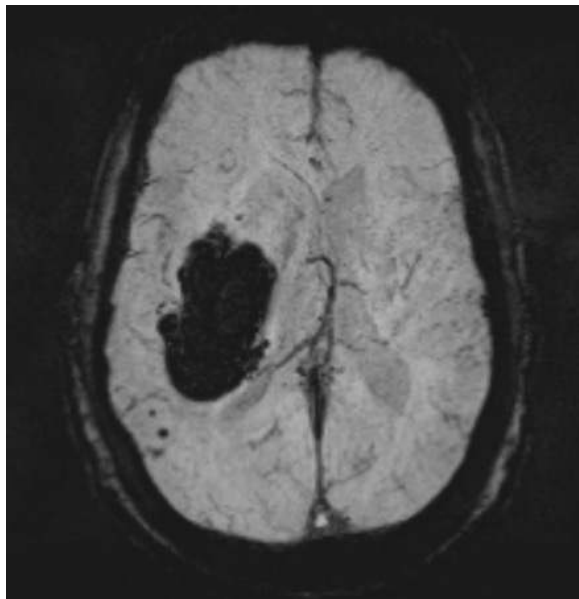
“**W**hite” on DWI & “**D**ark” on ADC = “restricted diffusion”
= acute stroke <14d old



Approach to MRI Brain

GRE or SWI

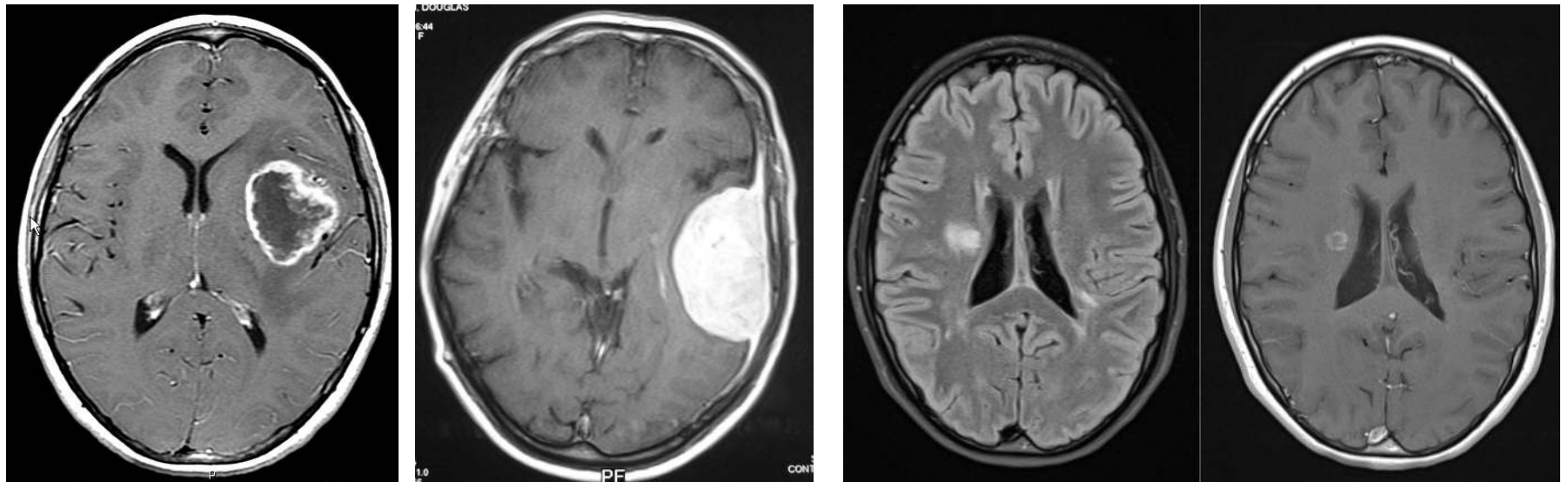
blood of any age (or calcium)



Approach to MRI Brain

T1 with GAD

enhancement = tumour, new demyelinating lesion, infection etc



COVID & Neurology

Association between covid-19 vaccination, SARS-CoV-2 infection, and risk of immune mediated neurological events: population based cohort and self-controlled case series analysis

BMJ 2022 ; 376 doi: <https://doi.org/10.1136/bmj-2021-068373> (Published 16 March 2022)

Cite this as: *BMJ* 2022;376:e068373

- looked at rates of immune-mediated neurological events (GBS, ADEM, Bells Palsy, transverse myelitis) after vaccination and after COVID infection:
- RESULT:
 - no increase in immune-mediated neurological events after vaccination
 - increased risk of Bell's palsy, GBS, ADEM after COVID infection

BONUS Stroke/Imaging MCQ 1

59M with a history of dyslipidemia, hypertension, and pacemaker for bradycardia presents with 18h history of headache and sudden onset ataxia. CT head reveals a subtle hypodensity suggesting an acute ischemic stroke. He is started on aspirin and clopidogrel. What is the best imaging test to perform next?

- a) Carotid Dopplers
- b) CT angiogram aortic arch to vertex
- c) MRI angiogram aortic arch to vertex
- d) Conventional angiogram to assess for vasculitis
- e) chest x-ray

BONUS MCQ 2

A 78yF with poorly controlled diabetes is being treated for breast cancer with adjuvant chemo and radiation. She develops new low back pain and leg pain. Exam of the legs shows fasciculations in the quadriceps and gastrocnemius bilateral, hyporeflexia, and bilateral leg weakness 4-/5. Pain is worse with straight leg raise. What is the most likely diagnosis:

- a) cervical spine stenosis
- b) L5 disc protrusion
- c) Diabetic lumbar plexopathy
- d) Epidural metastasis

BONUS MCQ 3

A 45yM Sommelier is diagnosed with early Parkinson's Disease and has predominantly mild bradykinesia and mild rigidity on exam, minimal tremor. He is counselled about treatment options and is very concerned regarding side effects of medications such as dyskinesias due to his profession. Which is the best option for treatment?

- a) ropinirole
- b) trihexyphenidyl
- c) levodopa/carbidopa
- d) selegiline

BONUS MCQ 4

An 80y right handed F with recent PCI 6 days ago for NSTEMI presents with acute onset aphasia and right hemiparesis, neglect, facial droop. She was last seen well 4 hours ago at 7am. BP 178/102 Cr 152, INR 1.2, Plt 148.

Meds ASA 325mg/Plavix75mg /Metoprolol/crestor/ramipril, blister pack shows meds last taken this morning. There is no hemorrhage on CT head. There are no acute ischemic changes. What is the next best treatment.

- a) IV labetalol
- b) ASA po/pr
- c) IV tPA
- d) CTA

MCQ 5

45F recently returned from Japan presents after an episode of transient left facial droop and L arm weakness with left neglect lasting 30 minutes. BP is 108/66. She was started on dual antiplatelets. Over the two weeks she completes her stroke investigations. MRI Head was normal. CTA shows no significant atherosclerosis. Echo shows a 2.5cm PFO. Leg dopplers show no DVT. A 14d holter is unremarkable. LDL is 2.1, HbA1c is 5.2%. APLA testing is negative. What is the next step in management?

- A) complete 21d of ASA & clopidogrel and refer for PFO closure
- B) complete 21d of ASA & clopidogrel and atorvastatin 80mg
- C) Atorvastatin 80mg and refer for PFO closure
- D) ASA and refer for PFO closure
- E) ASA and clopidogrel x 3 months and atorvastatin 80mg

Bonus MCQ 6

A 78yF right handed with history of nonvalvular atrial fibrillation presents with 4 day history of confusion. She shows left sided neglect and could not remember how to get dressed or buckle up seatbelt to come to appointment. CT head shows subacute wedge-shaped R MCA infarct. BP 110/70, wt 49kg. Hgb 120, Cr 154. LDL 1.78. CTA shows no intracranial or carotid stenosis.

She was recently diagnosed trigeminal neuralgia and started on carbamazepine for pain.

Meds: apixaban 2.5mg bid, rosuvastatin 10mg daily, carbamazepine 250 mg bid

With respect to meds for secondary stroke prevention, what do you recommend?

- a) Increase apixaban to 5mg bid
- b) Increase rosuvastatin to 20mg daily
- c) Hold apixaban, treat with ASA 81 mg daily for two weeks, then resume apixaban 5 mg bid
- d) Stop carbamazepine

MCQ 7

2026

A 38F presents with 3 days of diplopia. ESR/CRP are normal. HbA1c is 5.4%. CT head with CT angiogram is unremarkable. Six months prior, she had 2 weeks of right leg numbness that resolved. MRI Brain currently shows a small enhancing pontine lesion near the midline. What is the next step in management?

- a) ASA 81mg and Clopidogrel 75mg x 3 weeks then ASA alone
- b) ECHO to assess for PFO
- c) IVIG 2g/kg divided over 5 days
- d) Prednisone 1mg/kg PO x 5 days
- e) Methylprednisilone 1g IV x 5 days

MCQ 8

71M with a history of schizophrenia and many years of paliperidone (and prior to that, quetiapine) use presents with one year of falls. He reports urinary incontinence, erectile dysfunction and has had several ER visits for syncope. On exam he has rigidity of both arms, drooling, and dysmetria with finger-to-nose and an ataxic gait. What is the most likely diagnosis?

- a) drug-induced parkinsonism
- b) progressive supranuclear palsy
- c) multi-system atrophy
- d) idiopathic Parkinson disease
- e) normal pressure hydrocephalus

MCQ 9

64M presents with 7 days of tingling in the feet and hands followed by weakness of legs, falls, difficulty opening jars and now has one day of drooling and bifacial weakness. He reports lower back and bilateral leg pain. Exam shows weakness of both legs and hands. Reflexes are 1+ in the biceps and triceps, absent at the brachioradialis and in the legs. Sensation to pinprick is slightly reduced in the feet. Vibration is decreased at the toes and ankles. He had a possible viral URTI 3 weeks ago but recovered fully. What is the most likely etiology?

- a) diabetic polyneuropathy
- b) granulomatosis with polyangiitis
- c) Guillain-Barre Syndrome
- d) transverse myelitis
- e) cauda equina syndrome

Bonus MCQ 10 2025

A 77yM presents to the ED w nausea, vomiting and dizziness. He was recently started on levodopa/carbidopa for tremor thought to be related to Parkinson's. He is waiting for a neuro consult.

On exam

BP 120/80 supine, 90/60 standing.

Tremor of both hands outstretched. Head bob and vocal tremor noted. Normal gait and remainder of neuro exam.

Which of the following is true?

- a) Dementia will develop within 1 year
- b) The patient's orthostasis is likely due to Parkinson's
- c) Propranolol is the first line treatment for his disease
- d) His picture is consistent with MSA.

Bonus MCQ 11 2025

A 61yM living w diabetes comes with complaints of numbness in both feet, and the occasional “stumble” while walking.

PMH: Diabetes, controlled on metformin X 8 years, Hypertension, controlled on ramipril 5 mg po daily.

Your examination reveals bilateral decrease vibration sense to the knee, and reduced proprioception at the ankle. A 10g monofilament screen for diabetic neuropathy is positive in both feet. Power is reduced to 4+/5 on hip flexion, 4/5 knee, plantar, dorsi-flexion. There is some increased tone in the lower extremities, and brisk patellar reflexes. Ankle jerks also brisk bilaterally.

Which of the following is the next best test to diagnose the cause of his findings?

- a) HbA1C
- b) TSH
- c) Vitamin B12
- d) MRI L-spine – rule out spinal stenosis
- e) EMG/NCS to assess for distal symmetric polyneuropathy

Bonus MCQ 12 2025

A 27yF presents to your community hospital with visual disturbances (I couldn't see the soap dispenser in my shower!) and paresthesias that developed after a 12/10 headache. She suffers from migraines and reports this is the worst headache of her life, not responsive to usual treatments.

PMH: Migraines, obesity BMI 33, G1P1

Meds: desogestrel/ethinyl estradiol 28 pack 1 per day, sumatriptan 100 mg po prn migraine, ibuprofen 400 mg po prn migraine

Physical Exam: BP 150/98. wt 100 kg. Severe photophobia refusing fundoscopic examination complaining of severe retroorbital pain. She is in obvious distress. There is visual and sensory left hemineglect, gaze preference to right and hemisensory loss on left.

CT head with venous contrast suggests a right dural venous sinus thrombosis.

Which of the following are NOT a part of your initial management?

- a) Prophylactic levetiracetam 750 mg po bid
- b) Dalteparin 10,000 u sc BID
- c) Admit to monitored bed and discuss with nearest neuroICU/Neurosurgery
- d) Acetazolamide 500 mg BID

Bonus MCQ 13 2026

A 53M was riding his motorcycle when he was involved in a collision. He was helmeted. He was taken to a trauma centre for rapid evaluation and found to have a right radius fracture. The following day, he developed a subtle right facial droop, right arm drift with weakness of the right hand and somewhat effortful speech which he found frustrating. MRI Head confirmed an acute stroke. Carotid dopplers were normal. What test is most likely to determine the etiology of the stroke?

- a) ECHO
- b) CT Angiogram
- c) Telemetry/Holter monitoring
- d) HbA1c and Lipid profile
- e) APLA testing

Bonus MCQ #14 2025

A 46yF Right handed presents with 7h history of left sided neglect, confusion and left homonymous hemianopsia. Initial CT head shows no stroke. She is admitted for workup and rehab. ECHO bubble study normal, ECG and telemetry no afib after 72h admission. CTA shows 30% stenosis of L ICA at bifurcation. What is the best next step in her workup to determine etiology of stroke?

- a) MR angiography
- b) Urine toxicology
- c) APLA testing
- d) 14 d holter

Bonus MCQ #15 2025

- 36F presents with vertigo and horizontal diplopia over the last 3 days. The symptoms are constant and steadily worsening. Exam shows that when attempting to look to the right, the left eye cannot adduct and the right eye has a right beating nystagmus. What is the likeliest etiology?
 - a) pontine tumour
 - b) myasthenia gravis
 - c) multiple sclerosis
 - d) pontine lacunar infarct
 - e) left midbrain infarct

Bonus MCQ #16 – 2025

A 63yo patient presents with parkinsonism including rigidity and bradykinesia. Which of the following findings would not suggest a Parkinson plus syndrome?

- a) asymmetry on exam
- b) falls within the first year of symptom onset
- c) syncopal events
- d) impairment of vertical saccades
- e) unilateral pyramidal findings
- f) ataxia on exam

Bonus Answers.

1) B CT angiogram aortic arch to vertex

(not carotid dopplers because this is likely cerebellar, not MRI because CTA is better for vessels and he has pacemaker, not conventional angio because vasculitis is exceedingly unlikely)

2) d) LMN disease - not (a). L5 disease would not explain the proximal weakness and fascic. Diabetic lumbar plexopathy is usu. unilateral. Favour metastatic disease assuming it is in LUMBAR spine causing cauda equina

3) C - ropinorole is a dopamine agonist so has same risk of dyskinesia and is sedation so wouldn't be ideal.

Trihexyphenidyl is mainly for tremors so wouldn't be first choice. We used to think the longer you are on L-Dopa, the more likely to get dyskinesias. However, it is more likely that the longer you have PD, more likely to get dyskinesia from L-Dopa (correlation, not causation). Additionally, he drinks wine for a living, so risk of hypertensive crisis if tyramine rich diet.

4) The correct answer is C. BP is at acute target <180/105 no need for labetalol. Pt took ASA, plavix this morning. There's no way she can chew 160mg ASA w L MCA syndrome. Giving TPA priority. DAPT is not a Contraindication. Recent arterial puncture will be either femoral or radial so not a contraindication. You're barely within the window. Give TPA without delay. You will have to hold anti platelet 24h so obviously high risk but alternative is certain major disability. She may still get a CTA to see if EVT candidate but as next best TREATMENT give TPA.

Bonus Answers.

5) Answer d). -stroke in young so PFO is possible cause especially as other investigations negative

-no dual antiplatelet as this appears to be a cardioembolic phenomenon (from PFO)

- LDL is not perfect, but in young pt with no athero on CTA, this is unlikely to be the main issue

6) Answer D is correct – stop carbamazepine - She likely had a drug-drug interaction as Carbamazepine will REDUCE doac levels making it less effective. Pt does not qualify for high dose apixaban based upon weight and CrCl. LDL is at target, this stroke not atheroembolic. No real guideline here - the fact she is >4 days into stroke and hasn't had ICH would suggest you can continue her apixaban here; if very large stroke would likely hold DOAC acutely - but either way this is 'wrong' as it increases dose of apixaban to 5bid (see above)

7) Answer E – diagnosis is MS flare given the enhancing lesion (probably MLF lesion causing INO and diplopia) and thus treatment is 1g IV methylpred x 5 days. Oral prednisone is an option but dose would be 1250mg PO x 5d (not 1mg/kg). IVIG is an option for MS flares, but not typical first treatment.

8) Answer C Early falls, symmetrical parkinsonism= parky plus... Incontinence, erectile dysfunction, orthostasis and cerebellar features are hallmarks of MSA. Drug-induced is second most likely, but would not cause autonomic and cerebellar features. NPH can cause falls and incontinence but not the rigidity/parkinsonism

9) answer c), this is a non-length dependant, subacute process, diminished reflexes, with recent viral illness suggestive of GBS

- the presence of facial weakness eliminates spinal cord or cauda syndromes

Bonus Answers

10) Propranolol is the first line treatment for his disease- they are describing essential tremor. Side effects of L-dopa include GI intolerance, orthostasis and dizziness.

11) C: B12 Deficiency can be acquired in chronic metformin users, and should be suspected with this exam. if B12 is low-normal, can also diagnose based upon MMA and homocysteine level. Metformin impairs absorption of the B12-Intrinsic Factor complex.

12) A – no role for prophylactic anti-seizure drugs. Per Stroke best practices acetazolamide can be considered for headache post CVT or for papilledema due to same.

13) Answer B – this is most likely a left carotid artery dissection causing a left MCA cortical stroke (face and arm weakness and broca-type aphasia). The normal carotid dopplers does not rule out a dissection.

Bonus Answers

14) **D – 14 d holter.** Intentionally terrible question to reflect those you email to us each year. This patient has been admitted 72h so urine tox not particularly high yield. They have a R MCA syndrome, and if forced to pick ONE test 14d holter highest yield as cardioembolic causes of stroke far more common than APLAS.

15) **Answer is c).** Young female with a left INO. Though a pontine stroke is possible, MS would be much more likely in a pt of this age. If this was a 65yo with vascular risk factors, pontine infarct would have been the best answer.]

16) **Answer is a).** Asymmetry is a key feature in idiopathic Parkinson disease. The disease starts on one side and is always a bit worse on that side in PD. Early falls are a hallmark of parky plus diseases. Syncope suggests dysautonomia in MSA. Impaired vertical saccades suggests PSP. Unilateral pyramidal findings and ataxia can be seen in MSA.

GOOD LUCK! YOU'VE GOT THIS!